Version November 2, 2026
Preprint typeset using LaTeX style openjournal v. 09/06/15

GENERATIVE AI FOR IMAGE RECONSTRUCTION IN INTENSITY INTERFEROMETRY: A FIRST ATTEMPT

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(Dated:)
Version November 2, 2026

ABSTRACT

In the last few years Intensity Interferometry (II) has made significant strides in achieving high-precision resolution of stellar objects at optical wavelengths. Despite these advancements, phase retrieval remains a major challenge due to the nature of photon correlation. This paper explores the application of a conditional Generative Adversarial Network (cGAN) to tackle the problem of image reconstruction in II. This method successfully reconstructs the shape, size, and brightness distribution of simulated, fast-rotating stars based on sparsely sampled spatial power spectra obtained by using two different hypothetical ground-based II facilities composed of six and nine Imaging Atmospheric Cherenkov Telescopes (IACTs), respectively. **Although the relatively simple synthetic example considered here could also be addressed using conventional parameter-fitting techniques, the results demonstrate the potential of machine-learning-based approaches for image reconstruction from II measurements. With the increased number of telescopes and improved baseline coverage expected from future facilities such as Cherenkov Telescope Array Observatory (CTAO), these approaches may provide a useful framework for investigating more complex stellar systems exhibiting diverse surface features. This potential motivates further examination of machine-learning techniques for II image reconstruction.**

Keywords: Intensity Interferometry, Phase Reconstruction, Generative Adversarial Network (cGAN), Imaging Atmospheric Cherenkov Telescopes

1. INTRODUCTION

Humans instinctively feel a symbiotic relationship with the stars. One of the primary scientific projects of humanity is to figure out what the stars are and how do they do what they do. The first obvious step in this project, beyond measuring their global parameters like diameter, mass, orbital and astrometric elements, is to obtain images of the stars with all details of the stellar surfaces. In case of the Sun, this is done routinely by observatories and Sun-observation satellites. But such a routine still remains a challenge even for the α-Centauri system, our nearest stellar neighbour. The objective is to achieve capability of high fidelity image reconstruction of distant stars. Two interferometry based techniques, namely, Michelson Interferometry (MI) and Intensity Interferometry (II) have emerged during the last century to address this objective. A discussion of the development of these two approaches and comparison of their respective merits and challenges can be found in, for example, Rai et al. (2025). The work reported here presents the results of a first effort at applying a Conditional Generative Adversarial (neural) Network (cGAN) to image reconstruction of a fast rotator using its simulated II observations.

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The foundational basis of II stems from the pioneering experiments and theoretical investigations carried out initially by Hanbury Brown and Twiss (HBT) (Hanbury Brown & Twiss 1956; Brown & Twiss 1957, 1958) and, later, by Glauber (1963) and others. The correlation between photons emitted by a thermal source within a certain bandwidth (known as "photon bunching", also widely referred to as the "HBT effect") measured by a pair of photon detectors in two partially coherent beams of light was reported in 1956 by Hanbury Brown & Twiss (1956). Rai et al. (2025) present a recent variant of this experiment, carried out with pseudo-thermal light. The HBT effect and the related theoretical investigations laid the foundation for the modern field of Quantum Optics.

Hanbury Brown and his collaborators led the creation and installation of the historic II facility at Narrabri, Australia, and reported the measurement of angular diameters of 32 stars and a few multiple star-systems (Hanbury Brown et al. 1974). Soon after this work, however, II observations of stars was stalled for over four decades due to the limits of the then available photon detectors and data processing equipment. With gradual mitigation of such issues, proposals to utilize Imaging Atmospheric Cherenkov Telescope (IACT) facilities for conducting II observations of stars have emerged (Le Bohec 2006; Nuñez et al. 2010; Nuñez et al. 2012a; Dravins et al. 2013) as a secondary science application of these facilities during moonlit nights. SII observations at VERI-

TAS, MAGIC, and H.E.S.S. are now being reported (e.g., Acharyya et al. 2024; Abe et al. 2024; Vogel et al. 2025). This approach has the potential to enhance the scientific output of existing IACT facilities and the upcoming ones such as the Cherenkov Telescope Array Observatory (CTAO). Simulations (e.g., Rai et al. 2021, 2022) have shown that recent advancements in photon detectors hold promising prospects of high-precision stellar parameter measurements by such facilities.

The thrust of such efforts has been to measure average and global parameters such as angular diameters, binary separations, and orbital characteristics of stars and star systems. High-fidelity imagery, however, would transcend such integrated view of the systems and offer direct insights into dynamic stellar surface phenomena, including limb darkening, convection cells, granulation, star spots, oblateness and gravity darkening in rapid rotators and atmospheric structures, akin to the detailed observations routinely conducted on our own Sun.

As it stands today, studies grappling with various issues of image reconstruction are being reported (Haubois et al. 2009; Norris et al. 2021; Liu et al. 2024, 2025). Projects of MI-based image reconstruction have made substantial progress as demonstrated in the cases of Betelgeuse (Haubois et al. 2009) and AZ Cyg (Norris et al. 2021). On the other hand, II-based methods are at a nascent stage. Recently Liu et al. (2024, 2025) have demonstrated through outdoor experiments using "active optical intensity interferometry", imaging of millimeter-scale targets located at km scale distances and achieving resolution 14 times better than a single telescope's diffraction limit. A "flexible computational algorithm" reconstructs images from intensity correlations, overcoming atmospheric turbulence and optical imperfections.

We report here, the results of our attempt – the first of its kind – to construct the gravity-darkened sky-images of fast rotating stars consistent with their respective simulated ground-based II-observations using a cGAN neural network architecture. Image reconstruction of gravity-darkened fast-rotating stars has long been examined using various methods in MI (van Belle et al. 2001; Domiciano de Souza et al. 2003, 2005; McAlister et al. 2005; Monnier et al. 2007; Pedretti et al. 2009; Zhao et al. 2009; Martinez et al. 2021). Recently photosphere oblateness of γ-Cassiopeia (Archer et al. 2025) has been measured at the VERITAS observatory using II. These results put our work in context, and our work presented here is a natural next step especially of the work by Archer et al. (2025). We implement a cGAN model (Isola et al. 2017) to reconstruct images of **projected elliptical stellar models** using their simulated Intensity Interferograms and sky-intensity distributions as input data for training, testing, and validation. The intensity interferograms are synthetically generated by simulations of the fast rotators using two hypothetical IACT facilities having, respectively, arrays of $N_T = 6$ and $N_T = 9$ Cherenkov telescopes. The images predicted by the trained cGAN shows promising results in reconstructing the stars' shapes and sizes. The reconstructed brightness distributions are then assessed using moments.

This paper is organized as follows. **Section 2 briefly discusses previous efforts at image reconstruction relevant to II. Section 3 discusses the signal and noise characteristics of II observations of fast-rotating stars (FRSs) using IACT facilities with N_T telescopes and presents the method used to incorporate the effect of Earth's rotation during the observation. Section 4 introduces the cGAN formulation and its structure. Section 5 details the parameter selection for training the network for the task of image reconstruction. Section 6 presents the results of the trained network, both visually and through image moments. Finally, Section 7 concludes the paper with a discussion of the overall results.**

2. PAST EFFORTS AT IMAGE RECONSTRUCTION RELEVANT TO INTENSITY INTERFEROMETRY

Several approaches have been developed for phase reconstruction in II. Gamo (1963) introduced triple-intensity correlation, applied by Goldberger et al. (1963) to microscopic systems and extended by Sato et al. (1978, 1979, 1981) to measure stellar diameters and phases. The results of this work were limited by low signal-to-noise ratio (SNR). Gerchberg (1972) proposed an iterative phase retrieval method using image and diffraction plane data. This approach was sensitive to initial estimates and had to struggle with convergence speed issues. Fienup (1982) improved this approach with a Hybrid Input-Output algorithm, enhancing robustness in noisy conditions. Holmes et al. (2010) utilized Cauchy-Riemann relations for 1-D and 2-D image reconstruction, applied to simulated stellar data with Imaging Cherenkov Telescope Arrays (Nuñez et al. 2010; Nuñez et al. 2012a,b), but faced computational complexity for higher dimensions. Li et al. (2014) introduced a regularized iterative method incorporating priors (e.g., sparsity) to mitigate noise and ill-posedness, though challenged by parameter tuning and initial guess sensitivity. The Transport-of-Intensity Equation (TIE), proposed by Teague (1983), retrieved phase from intensity variations across the planes. Zhang et al. (2020) solved TIE as a Poisson equation using a maximum intensity assumption, while Kirisits et al. (2024) combined TIE with the Transport of Phase Equation for improved accuracy across arbitrary apertures and non-uniform illumination, with convergence dependent on initial guesses and boundary conditions.

With non-linearity built into their architecture, artificial neural networks (ANNs) empowered by deep learning methods present a promising machine-learning based method for exploring the task of reconstructing images of stellar objects from ground-based observations. Convolutional Neural Networks (CNNs), with their specialized architecture for processing two-dimensional datasets, are a natural choice for image processing tasks. In astronomical image reconstruction projects, a common challenge is that the data from the interferometric plane are typically sparsely sampled and are noisy as well. Therefore, the CNN architectures and the deep learning methods employed must be capable of reliably learning both the global context of the training dataset and the local features within it. Among the various CNN architectures, U-Net models (Ronneberger et al. 2015) have proven successful in such tasks.

Furthermore, given that achieving a high signal-to-noise ratio (SNR) is often challenging in astronomical datasets, it is immensely beneficial if additional data can

be generated using the available information from the observed sky density distribution and ground-based observations (II data, in our case) of the sources under investigation. Generative Adversarial Networks (GANs), introduced by Goodfellow et al. (2014), have been successful in such data augmentation tasks. Conditional GAN (cGAN) architectures, proposed by Mirza & Osindero (2014) and applied to a wide variety of datasets by Isola et al. (2017), leverage additional information about the images in the training datasets and have demonstrated remarkable robustness in image recovery across diverse data types.

In the astrophysical context, Schawinski et al. (2017) employed a GAN model to recover features, such as spiral arms, central bulges, and disk structures of galaxies, from noise-affected images. Mustafa et al. (2019) developed and customized a Deep Convolutional GAN, dubbed "CosmoGAN", capable of generating high-fidelity weak-lensing convergence maps of dark matter distribution that statistically reproduce real weak lensing structures. Coccomini et al. (2021) have successfully generated credible images of planets, nebulae, and galaxies using "lightweight" and "physics-uninformed" GANs to produce synthetic images of celestial bodies. They also generated a "Hubble Deep Field-inspired" wide-view simulation of the universe.

Encouraged by the success of machine-learning approaches, especially GAN-based structures, across various areas of Astrophysics, we have attempted to adapt this technique to the task of image reconstruction in II.

3. INTENSITY INTERFEROMETRY WITH IACT ARRAYS

This section provides a conceptual overview of the signal and its signal-to-noise ratio (SNR), the role of telescope arrays in collecting the measurements, and the stellar model used to represent the observed source.

3.1. *The signal for Intensity Interferometry*

As a simple example, let us consider an identical pair of IACTs pointed at a star. Suppose the two telescopes simultaneously measure the intensity of radiation $I_1(t)$ and $I_2(t)$, respectively. The signals from these detectors are cross-correlated and averaged over time, yielding the second order ($n = 2$) correlation of these intensities as (cf. Acciari et al. 2020; Dravins et al. 2013)

$$g_{12}^{(2)} = \frac{\langle I_1(t) \cdot I_2(t+\tau) \rangle}{\langle I_1(t) \rangle \cdot \langle I_2(t) \rangle} \tag{1}$$

where τ is the time delay between the telescopes. For spatially coherent and randomly polarized light, Eq. (1) reduces to the relation (sometimes called the Siegert relation, see e.g., Acciari et al. 2020).

$$g_{12}^{(2)} = 1 + \frac{\Delta f}{\Delta \nu} |V_{12}|^2 \tag{2}$$

where Δf is the electronic bandwidth of the photon detectors which measure the intensities and $\Delta\nu$ is the frequency bandwidth of the filters employed in the telescopes to observe the star. Values of $\Delta\nu \sim 1\,\mathrm{THz}$ and $\Delta f \sim 1\,\mathrm{GHz}$ are typical of recent work. In Eq. (2), V_{12}, referred to as the complex visibility function, is the Fourier transform of the source brightness distribution. It contains information about the star's angular diameter. However, the phase information is lost since we measure only the absolute value $|V_{12}|^2$. In observational astronomy, the correlation is often expressed in terms of the normalized contrast c_{12}, given by:

$$c_{12} = g_{12}^{(2)} - 1 = \frac{\Delta f}{\Delta \nu} |V_{12}|^2 \tag{3}$$

with $|V_{12}|^2$ being a function of baseline $b = \sqrt{u^2 + v^2}$ on the observational plane (u, v). This implies the strength of the signal would be enhanced if a larger number of baselines or pairs of telescopes are employed.

The total correlation obtained by all the pairs of N_T telescopes in an observation facility, excluding the zero-baseline cases, becomes

$$c_{N_T} = \sum_{\substack{i,j=1 \\ i<j}}^{N_T} c_{ij}. \tag{4}$$

3.2. *Signal-to-Noise Ratio for Intensity Interferometry*

The primary purpose of IACT arrays is to observe the Cherenkov showers generated in the upper atmosphere by high-energy gamma rays (with energy $E \geq 30$ GeV) arriving from cosmic sources. When employed for II observations, these telescopes focus light received from a sky source onto their respective set of photo-multiplier tubes (PMTs), see e.g., (Aleksić et al. 2016) with appropriate specifications. In the simulation models adopted here, we consider, in succession, two arrays of IACTs each with similar properties. The positional configurations of these arrays of $N_T = 6$ and $N_T = 9$ IACTs are respectively shown in the left and the right panels of Fig. 1. The optical signal directed to a PMT is filtered using a spectral filter with a chosen mean observational wavelength λ and corresponding bandpass $\Delta\lambda$. The use of filters not only reduces background noise but also improves the signal quality and the efficiency of the PMTs. Filtering background skylight becomes even more significant in II observations, as, currently, these are carried out during bright moon-lit nights when the primary function of the IACTs (of observing Cherenkov Showers) is rendered infeasible. It is important to note that the light from the stellar source is focused on a PMT attached with each of the telescopes during II observations.

The significance of the signal can be expressed in terms of the signal-to-noise ratio (SNR), which depends on many factors. However, most importantly, it does not depend on the optical bandwidth $\Delta\nu$ of the radiation for a two-telescope correlation. The explanation for the independence of the SNR from $\Delta\nu$ is provided in several works (e.g., subsection 4.1 of Rai et al. 2021). The Signal-to-Noise Ratio (SNR_{12}) produced by the considered identical pair of IACTs becomes

$$\mathrm{SNR}_{12} = A \cdot \alpha \cdot q \cdot n \cdot F^{-1} \cdot \sigma \cdot \sqrt{\frac{T\Delta f}{2}} \cdot |V_{12}|^2 \tag{5}$$

Here, A is the total mirror area, α is the quantum efficiency of the PMTs, q is the throughput of the remaining optics, and n is the differential photon flux from the source. The excess noise factor of the PMTs is represented by F, T denotes the observation time, and σ is

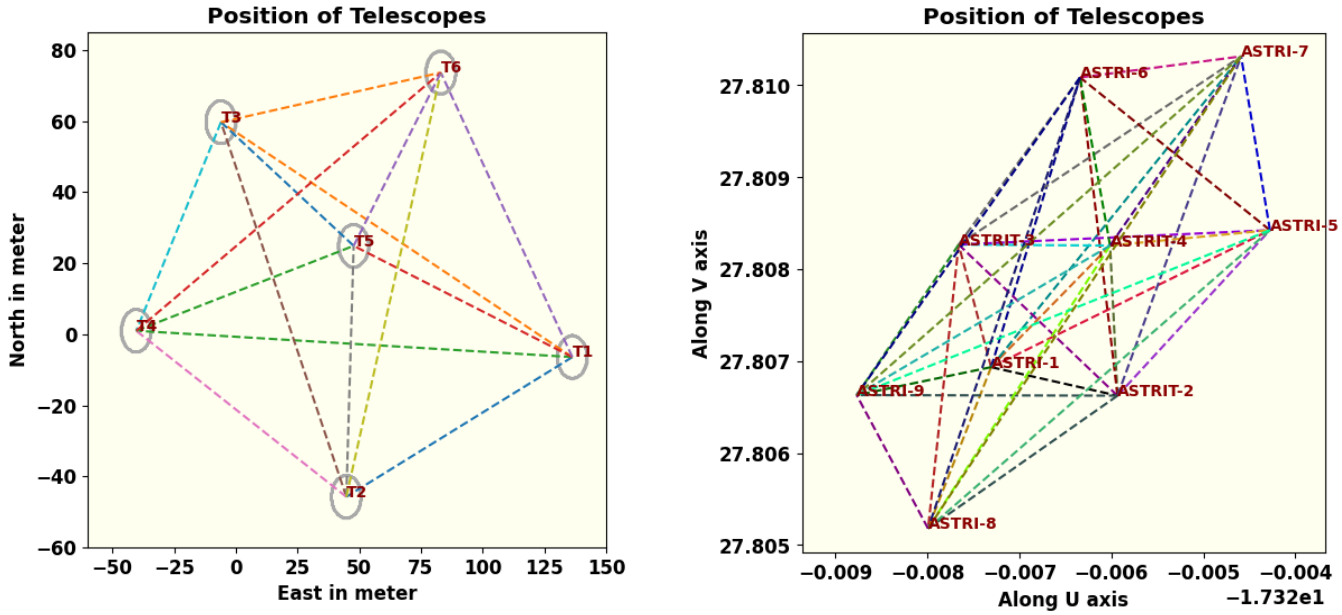


FIG. 1.— A schematic representation of a hypothetical observation facility with an array of $N_T = 6$ Imaging Atmospheric Cherenkov Telescopes (IACTs) (left panel) and $N_T = 9$ IACTs that mimic the telescope configuration of ASTRI (right panel Scuderi et al. (2022)). In the work presented here, these two sets of arrays are used in simulating the II observation of the fast rotators.

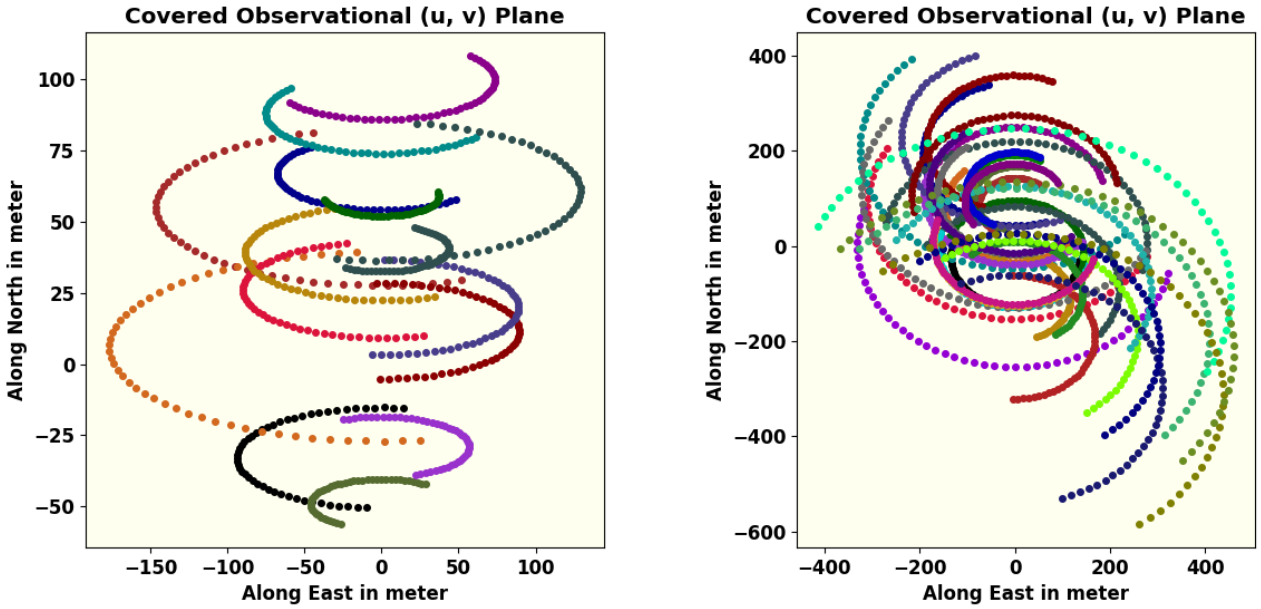


FIG. 2.— The tracks of the baselines of the two hypothetical II observation facilities shown in fig. 1 generated (synthetically) due to Earth's rotation during one night of simulated II observation. The number of baselines scales as square of number of telescopes in the array thus leading to greater coverage of the observational plane and better image reconstruction prospects.

the normalized spectral distribution of the light (including filters) (see e.g., Acciari et al. 2020).

The total SNR obtained by all the pairs of N_T telescopes in an observation facility excluding the zero-baseline cases, becomes

$$\mathrm{SNR}_{N_T} = \sqrt{\sum_{\substack{i,j=1 \\ i<j}}^{N_T} (\mathrm{SNR}_{ij})^2}. \qquad (6)$$

Point to note in Eq. 5 and Eq. 6 is that all the N_T IACTs in the facility along with their associated optics and other parameters are considered to be identical.

Although this section derives the SNR for II, the present simulations do not explicitly use this expression to calibrate the noise amplitude or model baseline-dependent uncertainties in photon-correlation measurements. Instead, salt-and-pepper noise is introduced as a phe-

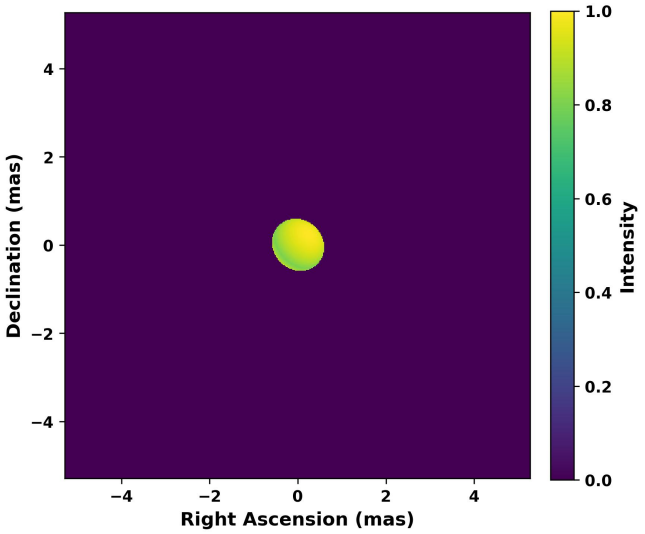


FIG. 3.— This figure shows one of the simulated fast rotating stars (FRS). Brightness is the highest at the poles; gravity darkening is visible along the equator. A total of 15730 such images of FRS with different parameter values have been generated to train the model.

nomenological perturbation to investigate the robustness of the proposed GAN-based reconstruction framework and its ability to recover stellar morphology from noisy image representations. Therefore, the adopted noise model should not be interpreted as a quantitative representation of observational noise in an actual II experiment. Rather, the simulations serve as a proof of concept for the proposed reconstruction approach under phenomenological perturbations. Incorporating noise calibrated to the II SNR and accounting for the statistical uncertainties of individual baseline measurements will be important directions for future work.

3.3. *Baseline considerations*

The measurement of the size of stellar objects via squared visibility depends on the distance between the telescopes, known as the baseline b.

$$|V_{12}(b)|^2 = \frac{c(b)}{c(0)} \tag{7}$$

For achieving a good SNR with a given telescope configuration, covering as much as possible of the interferometric plane is always desirable. If the source is at the zenith, the coordinates in the Fourier plane (u, v) are given by:

$$(u, v) = \frac{1}{\lambda}(b_E, b_N) \tag{8}$$

where b_E and b_N are, respectively, the baselines expressed in east and north coordinates. However, the sources can be anywhere on the sky, and the telescopes are stationary and may also have different relative altitudes b_A depending on the available terrain. In order to gather maximum possible information on the source during the observation session and to cover as much of the observational plane as possible during such sessions, Earth's rotation must be taken into account using rotated baselines. For a given stellar source with declination δ and hour-angle h, as observed by telescopes located at latitude l, Eq. 9 provides the rotated baselines for a given pair of telescopes (see e.g., eqs. 8–10 from Baumgartner et al. 2020) with the R-matrices representing the respective rotation operations.

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = R_x(\delta) \cdot R_y(h) \cdot R_x(-l) \begin{pmatrix} b_E \\ b_N \\ b_A \end{pmatrix} \tag{9}$$

Fig. 2 shows, respectively, the tracks of 15 and 36 baselines generated, due to Earth's rotation from the telescopes shown in the left and the right panel of Fig. 1. Since every pair of telescopes traces an ellipse in the Fourier plane, the total number of ellipses scales as

$$\mathcal{N} = \tfrac{1}{2} N_T \cdot (N_T - 1) \tag{10}$$

where N_T is the number of telescopes considered. As the number of baselines increases quadratically with N_T, II benefits greatly from a large number of telescopes. The upcoming CTAO Global facility with upto 65 IACTs across both the hemispheres can offer many more baselines and play a major role in both γ-ray observations and II. Dravins et al. (2013) has considered the telescope configurations being planned and has showed how it would provide a dense coverage of the interference plane.

3.4. *Synthetic stellar brightness distribution*

We construct synthetic stellar images using a projected elliptical model whose geometry is controlled by the stellar radius, inclination, position angle, and axis ratio. The position angle, p, specifies the rotation of the model in the image plane. The rotated coordinates are defined as

$$x' = x\cos p + y\sin p, \qquad y' = -x\sin p + y\cos p. \tag{11}$$

For a given inclination i and axis-ratio parameter s_q, the projected regions are constructed iteratively using a parameter $r_s \in [0, \pi]$. The r_s is the iteration parameter used to generate the sequence of regions; it is not the radial distance of an individual image pixel. At each iteration, the displacement parameter is

$$z(r_s) = s_q\, R\, \sin i\, \cos r_s, \tag{12}$$

where R denotes the stellar radius parameter. The pixels belonging to the corresponding projected region satisfy

$$x'^2 + \left(\frac{y' - z(r_s)}{\cos i}\right)^2 < [R \sin r_s]^2 . \tag{13}$$

The intensity assigned to these pixels is

$$B(r_s) = \frac{4 + |\cos r_s|}{5}. \tag{14}$$

Thus, Eq. 13 defines the spatial extent and displacement of each projected region, whereas Eq. 14 specifies its assigned brightness and it ranges from 0.8 to 1.0 in relative units.

The synthetic dataset is generated by varying four model parameters: stellar radius, inclination, position angle, and axis ratio. The stellar radius is sampled at 65 values between 3×10^{-9}

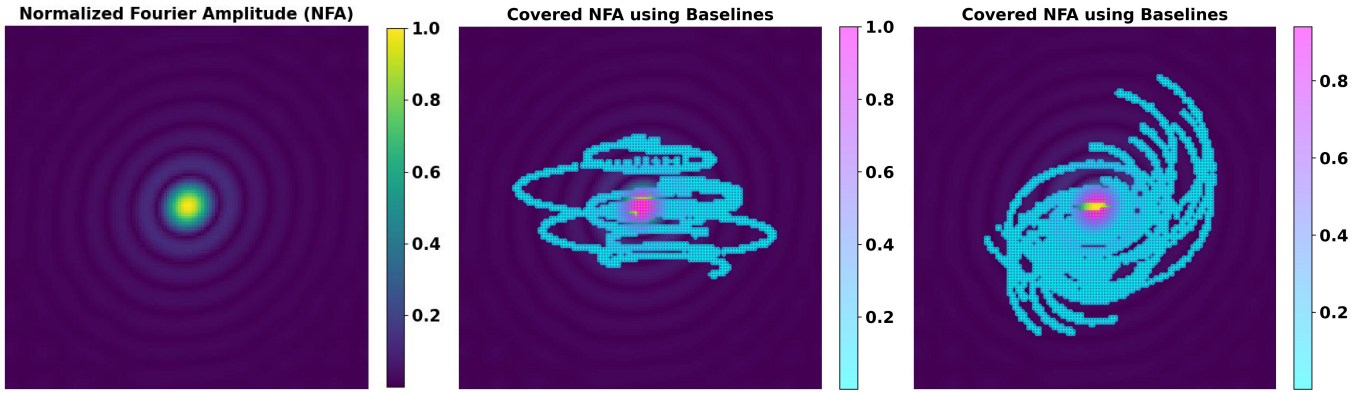


FIG. 4.— **The left panel shows the absolute value of the two-dimensional Fast Fourier Transform (FFT) of the source depicted in Fig. 3, displayed using the viridis colormap. It represents the interferometric (u, v) plane intensity distribution that would be generated by an infinite number of baselines (or an infinite number of telescopes) observing the source. The middle and right panels show the same full (u, v)-plane intensity distribution in the background using the viridis colormap, with the regions sampled by the baseline tracks illustrated in the left and right panels of Fig. 2, respectively, overlaid using the cool colormap. The middle and right** panels reflect the sparse nature of the signal received by a realistic finite number of telescopes and baselines sampled from the full (u, v) plane signal space. All figures are plotted on a linear scale and normalized to the maximum pixel value in each respective figure.

and 1.58×10^{-8}, with an increment of 2×10^{-10}. The inclination and position angle are each sampled at eight values over $[0, 2\pi)$, with an angular interval of $\pi/4$, while the axis ratio is varied over five values between 0.5 and 0.9 in increments of 0.1. For the two special inclinations, $i = 0$ and $i = \pi$, a single image is generated per stellar radius without varying the position angle or axis ratio. For the remaining six inclination values, all combinations of position angle and axis ratio are considered. Consequently, the dataset contains

$$N_{\rm images} = 65 \times [2 + (6 \times 8 \times 5)] = 15730 \qquad (15)$$

synthetic stellar images.

The adopted brightness prescription (Eq. 13) is a simplified, phenomenological representation of brightness variations associated with rapid rotation. It is not calculated directly from the local effective gravity, temperature, or emergent flux. Instead, it produces a smooth brightness distribution with enhanced intensity toward the projected polar regions and reduced intensity toward the equatorial region. Classical gravity-darkening models, such as those discussed by Von Zeipel (1924) and Lucy (1967), relate stellar surface properties to local effective gravity under specified physical assumptions. The adopted prescription (Eq. 13) is not intended to reproduce these models quantitatively. Accordingly, the dataset provides a controlled benchmark for evaluating cGAN-based reconstruction of stellar morphology and brightness distributions, rather than a validation of a physically self-consistent gravity-darkening model. The present models also omit starspots, granulation, detailed limb-darkening profiles, and tidal deformation. Incorporating more physically motivated intensity distributions and more complex photospheric structures, including those of interacting binary systems, is an important direction for future work.

3.5. *A Fictitious Fast Rotating Star: Our Test Case*

In our work presented here, we attempt image reconstruction of fast-rotating stars using their simulated II observation in a cGAN architecture. Fast-rotating stars are important test cases for understanding various astrophysical processes, including stellar evolution, internal structure, and dynamical behaviour over time. Fast rotation causes stars to adopt an oblate shape, flattening at the poles and bulging at the equator due to the stronger centrifugal force (e.g., Von Zeipel 1924; Maeder 1999). **Fig. 3 shows an image following Eq. 14 and qualitatively representing a fictitious fast-rotating star.** First direct interferometric detection of stellar photospheric oblateness (of Altair) was pioneered by van Belle et al. (2001) using the Palomar Testbed Interferometer (PTI) and the Navy Prototype Optical Interferometer (NPOI). Gravity darkening due to fast rotation was first observed through interferometric and spectroscopic data from CHARA Array for the fast-rotating star Regulus (McAlister et al. 2005). As pointed out earlier, these two pieces of work, all using Michelson Interferometry, make a subset of several others (van Belle et al. 2001; Domiciano de Souza et al. 2003; McAlister et al. 2005; Domiciano de Souza et al. 2005; Monnier et al. 2007; Pedretti et al. 2009; Zhao et al. 2009; Martinez et al. 2021). The first observation of photospheric oblateness (of γ Cassiopeiae or γ-Cas) using II has been recently carried out at VERITAS observatory and is reported by Archer et al. (2025). Reportage of such observations of other γ-Cas like targets and other class of FRS by Cherenkov Telescope arrays, such as the MAGIC array, are expected by 2026. In addition, observation and measurement of gravity darkening using II is the natural next step and is yet to be reported. In this context, our work of reconstructing the image of FRSs from their II-simulated observations using cGAN is an attempt at solving this inverse problem along with mitigation of loss of phase information in II.

II records the photons arriving at the telescopes from the stellar source. The correlation of photon arrival times between pairs of telescopes provides the squared visibility, as given by Eq. (7) and explained in subsec-

tion 3.1. **The two-dimensional FFT of the source image provides the Fourier-amplitude distribution over the spatial-frequency plane. To simulate the sparse sampling imposed by the available baseline configuration, we construct a binary mask from the baseline-track image. The normalized Fourier-amplitude distribution is then multiplied element-wise by this binary mask. Consequently, the resulting array retains the Fourier amplitudes at the pixel locations selected by the mask, while the remaining pixels are set to zero. The physical coordinates of the FFT pixels are assigned using the angular pixel scale of the original source image and the observing wavelength, allowing the sampled Fourier amplitudes to be associated with their corresponding spatial frequencies in the (u, v) plane.**

The left panel of Fig. 4 shows the normalized absolute value of the two-dimensional Fast Fourier Transform (FFT) of the source image shown in Fig. 3, representing the full intensity distribution in the interferometric (u, v) plane. **This distribution is displayed using the viridis colormap and represents the idealized coverage obtainable with an infinite number of baselines**. In practice, only a sparse sampling of the full (u, v) plane is accessible because of the finite number of telescopes and the limited observation time. The middle and right panels of Fig. 4 illustrate this limitation by overlaying the regions sampled by the available baseline tracks on the full intensity distribution. **The background full distribution is displayed using the viridis colormap, while the sampled regions are highlighted using the cool colormap**. These panels correspond to two hypothetical observation facilities comprising arrays of $N_T = 6$ and $N_T = 9$ imaging atmospheric Cherenkov telescopes (IACTs), with their relative positions determining the available baselines, as shown in the left and right panels of Fig. 1 and Fig. 2, respectively. **We simulate one night of intensity interferometry observations for each array and use the resulting sparsely sampled data to train a conditional generative adversarial network (cGAN) to reconstruct the source images.**

4. GENERATIVE ADVERSARIAL NETWORKS

Generative Adversarial Networks (GANs) were introduced by Goodfellow et al. (2014). A GAN model involves two competing deep neural network models, referred to as the Generator and the Discriminator. These two networks engage in a zero-sum "Minimax" game, as in Game Theory. Given a real data set $\{x_i\}$ (for example, a set of real images) drawn from some unknown distribution $P_{\rm data}(x)$ generated by some unknown or ill-understood process, the objective here is to generate a new set (of images) whose probability distribution should match $P_{\rm data}(x)$ as closely as possible. During the training of the two models, the Generator samples a latent variable z from a known prior distribution $P_z(z)$ (*e.g.*, the Normal Distribution) and produces a synthetic sample $G(z)$ to start with and, subsequently, based on updates received from the Discriminator as its training progresses. The Discriminator, being a probabilistic binary classifier, receives either a real data sample x or a generated sample $G(z)$, and outputs a probability that the input is real. The Discriminator aims to maximise its classification accuracy, while the Generator aims to fool it by trying to minimize it. The training of these two networks proceeds alternately leading to the optimization of the adversarial loss function $L(D, G)$ given by

$$\begin{aligned} L(D,G) &= \min_G \max_D V(D,G) \\ &= \mathbb{E}_{x\sim p_{\rm data}(x)}\left[\log D(x)\right] \\ &+ \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D(G(z))\right)\right]. \end{aligned} \tag{16}$$

The two neural networks $G(z) \equiv G(z;\theta_G)$ and $D(x) \equiv D(x;\theta_D)$ are parameterized by θ_i with $i = G$ or D respectively. During its training, the Generator samples random noise vector z from a Normal Distribution and generates the differentiable function $\hat{x} \equiv G(z)$ that maps z to the data space x. Through such maps the Generator generates its own distribution $p_{\rm G}(\hat{x})$ and through the training episodes, specifically by iteratively updating its parameters $\theta_{\rm G}$, aligns this distribution with the distribution of real data $p_{data}(x)$. The training data set provided to the Discriminator is constructed by mixing real data points x and generated data points $\hat{x}$ in equal ratio. The Discriminator generates the function $D(x)$ that represents the probability that x originates from real data. Eq.(16) implies that training of the GAN model moves towards maximization of the expectation of $D(x)$. Through this process, both the parameters $\theta_{\rm G}$ and $\theta_{\rm D}$ are optimized such that $p_{\rm G}(\hat{x})$ gets maximally aligned with $p_{data}(x)$.

For a given fixed Generator $G(z)$, the problem can be reformulated as:

$$\begin{aligned} \max_D V(D,G) &= \mathbb{E}_{x\sim p_{\rm data}}\left[\log D^*_G(x)\right] \\ &+ \mathbb{E}_{x\sim p_G}\left[\log\left(1 - D^*_G(x)\right)\right] \end{aligned} \tag{17}$$

where D^*_G denotes the optimum of the discriminator. As seen in Eq. 18, the global optimum of Eq. 17 is achieved if and only if $p_G = p_{\rm data}$. Furthermore, if both G and D are allowed to reach their respective optima, – the so called Nash point of the Minimax game – $p_{\rm G}$ converges to $p_{\rm data}$.

$$D^*_G(x) = \frac{p_{\rm data}(x)}{p_{\rm data}(x) + p_G(x_{gen})} \tag{18}$$

At this point, the Discriminator finds its job no better than random guessing. A more comprehensive discussion of the problem, including proofs, is provided in Goodfellow et al. (2014).

Subsequently, the GAN framework was extended to a conditional model (Mirza & Osindero 2014). This new model, known as "conditional GAN" or cGAN injects a conditioning variable y into both networks: the Generator now generates $G(z \mid y)$, and the Discriminator evaluates $D(x \mid y)$. The adversarial objective becomes

$$\begin{aligned} V(D,G) &= \mathbb{E}_{x,y\sim p_{\rm data}(x)}\left[\log D(x|y)\right] \\ &+ \mathbb{E}_{z,y\sim p_z(z)}\left[\log\left(1 - D(G(z|y)|y)\right)\right] \end{aligned} \tag{19}$$

The conditional variable y in a cGAN can be various additional information including images, labels or text contextual to "ground truth" real data x. Among various types of cGANs, Pix2Pix GAN with its image-to-image translation design is suitable for the task at hand.

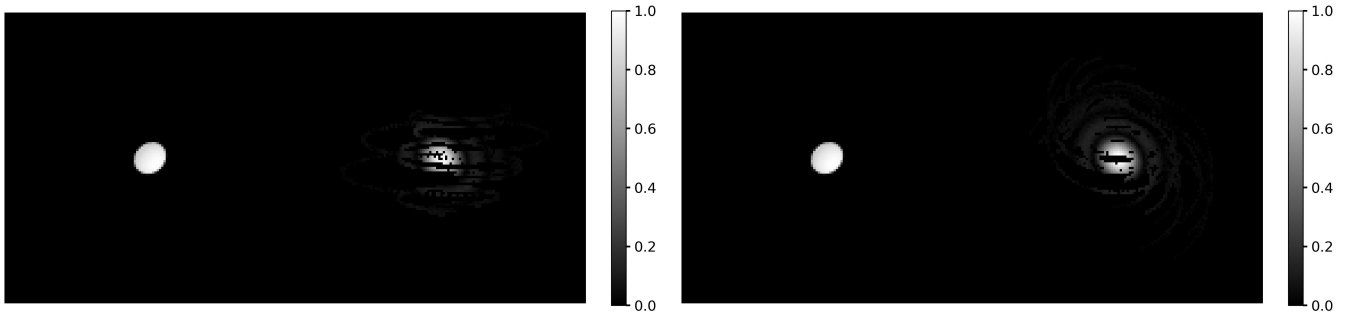


FIG. 5.— Two illustrative pieces of example of the input used for training the cGAN model. The picture on the left in both of the panels represents the source image in grey scale, which serves as the ground truth or the real data (x), as mentioned in the Flow Diagram (Fig.6 discussed later). The picture on the right, in both of the panels represents the simulated II observation pattern of the source image x. These two II observation patterns of the two panels are generated using the arrays of IACTs and the Earth's rotation. The pattern on the left (right) panel is generated using the array of $N_T = 6$ ($N_T = 9$) IACTs. This pattern referred to as y, in the Flow Diagram (Fig.6 discussed later) forms the "condition" during the training to which the GAN model has to conform. Salt-and-pepper noise is added to this pattern for enhancing the robustness of the cGAN model. Together, the source image x on the left and the II pattern y on the right form a training pair. The GAN learns to construct the "predicted" image similar to ground truth x from the noisy baseline signal y. The grayscale in both images is normalized to the brightest pixel.

In our work, the conditioning variable y represents the input signal derived from the simulated intensity-interferometry (II) measurements. The generator learns a mapping from this input signal to a reconstructed stellar image, while the discriminator evaluates the consistency of a candidate image with the corresponding input signal. Although the general cGAN formulation in Eq. 19 includes a latent noise vector z, our implementation does not explicitly supply such a vector to the generator. Instead, the generator directly maps the conditional input y to the reconstructed image, which we denote by $G(y)$. Thus, the model implemented in this work is a conditional GAN with a deterministic U-Net generator and a conditional PatchGAN discriminator.

Isola et al. (2017) further observed that combining the cGAN from Eq. 19 with the traditional $L_1(G)$ loss (also known as the Mean Absolute Error) improves the results, as the Generator is encouraged to produce outputs closer to the target image in a pixel-wise sense. We adopt this approach in the training of our cGAN model by including $L_1(G)$ defined in Eq. 20

$$L_1(G) = \mathbb{E}_{(x,y)\sim p_{\rm data}(x,y)}\left[\|x - G(y)\|_1\right]. \tag{20}$$

The total adversarial loss function, along with the L_1 loss modulated by a hyperparameter λ then becomes

$$L_{tot} = \min_G \max_D \left[V(D,G) + \lambda L_1(G)\right]. \tag{21}$$

This type of network has demonstrated remarkable robustness across a variety of applications. For example, it can generate colored images from grayscale inputs based on architectural labels, transform images from day to night, and even predict maps from satellite data. A more extensive list of applications is provided in Isola et al. (2017).

4.1. *Generator*

As discussed above, in a GAN the Generator, a deep neural network in **itself**, is responsible for producing synthetic data, in this case, images that resemble those of a fast-rotating star. **In this work, the generator is implemented as a U-Net convolutional network (Ronneberger et al. 2015). The network consists of an encoder-decoder architecture with skip connections between corresponding encoder and decoder levels. The encoder progressively reduces the spatial resolution while increasing the number of feature channels. In the implementation, the encoder contains seven down-sampling blocks with 64, 128, 256, 512, 512, 512, and 512 filters, respectively. The decoder contains six up-sampling blocks with 512, 512, 512, 256, 128, and 64 filters, followed by a final transposed convolution that produces the reconstructed image. The decoder feature maps are concatenated with the corresponding encoder feature maps, allowing spatial information from earlier layers to be retained during reconstruction.**

Dropout is enabled in the first two up-sampling blocks to help mitigate overfitting. The final layer uses a hyperbolic tangent (tanh) activation. Unlike the general latent-variable formulation of a cGAN, the implemented generator receives only the input signal y and produces the reconstructed image $G(y)$. The combination of multi-scale feature extraction and skip connections is intended to preserve spatial structure while learning the mapping from the input signal representation to the target stellar image.

During the training, the total Generator Loss function L_G including the L_1 loss defined in eq.(20) that is minimized is given by

$$\begin{aligned} L_G = & -\mathbb{E}_{(x,y)\sim p_{\rm data}(x,y)}\left[\log D(G(y) \mid y)\right] \\ & + \lambda\, \mathbb{E}_{(x,y)\sim p_{\rm data}(x,y)}\left[\|x - G(y)\|_1\right]. \end{aligned} \tag{22}$$

Here,

- x denotes the target stellar image corresponding to the input signal y.
- y denotes the simulated II input signal representation.
- $G(y)$ denotes the reconstructed stellar image produced by the generator.

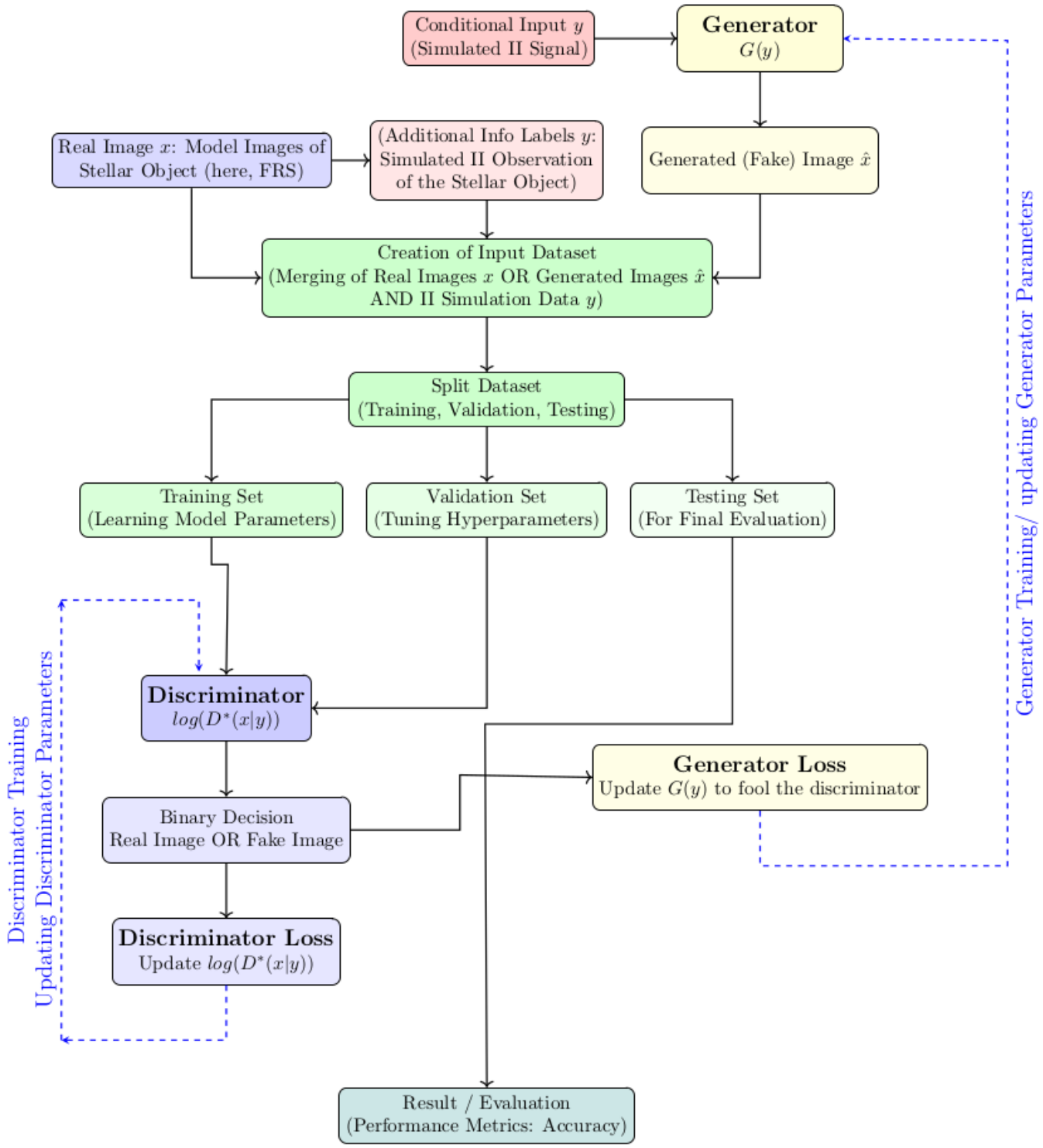


FIG. 6.— A schematic representation of the features of the cGAN model used in this work and the iterative process of its training, validation and testing. The process constitutes four broad stages: (1) choice of an appropriate GAN architecture including both the Discriminator and the Generator (not shown in this figure) (2) preparation of the Training, Validation and Testing datasets and (3) Training and Validation of the Model (4) Testing and Evaluation of the Model. The stages (2), (3) and (4) are depicted in this figure. The datasets are prepared in three broad steps: (i) generating the "ground truth" images x of fictitious fast rotating stars (FRS) , the sparse II observation signal y used as the "condition" images in the Model and the generating the ground-truth images x of fictitious fast-rotating stars and the corresponding sparse II input signals y, which serve as the conditioning inputs to the model, and finally (iii) partitioning the full dataset into Training Set, Validation Set and the Testing Set. After the iterative training of the Model and its validation process is complete ("Nash point" of the Minimax Game is reached), the Model is tested using the Testing Set and evaluated.

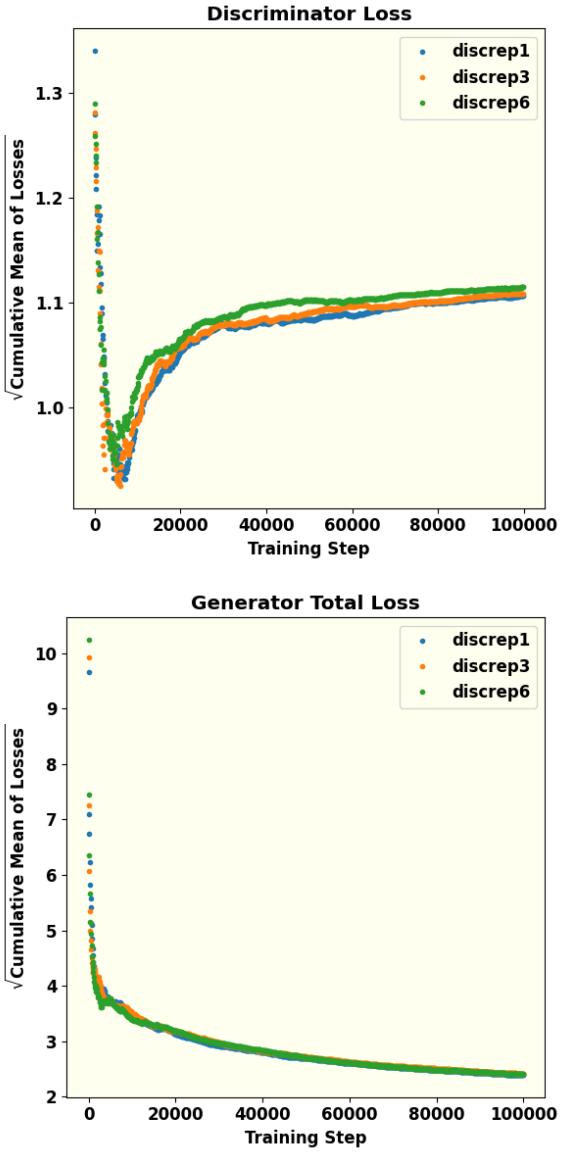


FIG. 7.— These figures show the effect of the ratio of episodes of Discriminator training per every episode of Generator training. This hyperparameter is termed Discriminator repetition or (discrep) in the figures. The square root of the cumulative mean of the losses are plotted against the training steps for 3 values of this ratio. Understandably, this ratio has a higher impact on the Discriminator loss than the Generator loss. Equal number of episodes of training produces minimum cumulative loss of the Discriminator. The dip in the Discriminator loss during the initial phases of its training can be interpreted as its early success in detecting "fake" (or generated) images because of a poorly trained Generator. With gradual training of the Generator, the success rate of the Discriminator decreases and eventually approaches saturation with equal probability of being successful in telling "fake" from real. The continual decrease and eventual saturation of the Generator loss is a result of its training to generate better images with increasing number of steps.

- $D(G(y) \mid y)$ denotes the discriminator output for a candidate image conditioned on y. In our implementation, this output is a grid of logits rather than calibrated probabilities.

- λ controls the relative contribution of the L_1 reconstruction loss.

4.2. *Discriminator*

The Discriminator is tasked with classifying the images produced by the Generator as either real or fake. It takes a real image from the dataset (often referred to as the target image for the Generator) and provides feedback to guide the Generator toward producing more accurate images. In this work, the PatchGAN model (Isola et al. 2017) is employed as the Discriminator. Unlike a traditional global classifier, PatchGAN evaluates individual patches of the image, outputting a grid of predictions rather than a single scalar value. Each element in the grid corresponds to the "realness" of one patch of the image under examination of the Discriminator at a time. The final loss of the Discriminator is the average over all the patch responses. Evaluating the "realness" of the input image in terms of its constituent patches facilitates capture of texture/ style and other high frequency components in the image. As compared to a global discriminator, it also reduces the number of parameters in the network thereby helping reduce computation cost. It also works on images with arbitrary sizes.

The discriminator follows a PatchGAN architecture (Isola et al. 2017) and receives the input signal together with a candidate stellar image, which may be either the ground-truth image or the generator output. In the implementation, salt-and-pepper perturbation is applied to the input signal before it is concatenated with the candidate image. The concatenated input is then processed by a sequence of convolutional down-sampling blocks. Subsequent layers use zero padding, convolution, batch normalization, and LeakyReLU activation, followed by a final convolution that produces a spatial grid of discriminator outputs. Each output corresponds to a local image region, allowing the discriminator to assess local characteristics of the candidate image in the context of the input signal. The discriminator is trained to distinguish real input target pairs from input generated image pairs.

The Discriminator Loss function L_D that is optimized during the training process is given by

$$\begin{aligned} L_D = & -\mathbb{E}_{(x,y)\sim p_{\text{data}}(x,y)} \left[\log D(x \mid y)\right] \\ & -\mathbb{E}_{(x,y)\sim p_{\text{data}}(x,y)} \left[\log\left(1 - D(G(y) \mid y)\right)\right]. \end{aligned} \tag{23}$$

where the arguments of the D and G functions are as noted in the text following Eq. (22). This loss is composed of two parts: one that assesses how accurately it identifies fake images (by comparing predictions to a target value of 0) and the other that measures how accurately the Discriminator identifies real images (by comparing predictions to a target value of 1).

The training procedure of these two components of the cGAN model can be outlined as follows:

- The discriminator D is updated by minimizing L_D, keeping G fixed (so that D becomes better at distinguishing real vs generated images).

- The generator G is updated by minimizing L_G, keeping D fixed, thus pushing G to generate images that are (a) judged "real" by D, and (b) close (in pixel-wise sense) to the ground-truth x.

5. NETWORK PARAMETERS

An appropriate cGAN architecture along with a set of hyperparameters was optimized by tuning. The objective of the model, as already mentioned, was to learn to faithfully reproduce a set of sky-images of fast rotators subject to the condition that those images are consistent with their simulated II observation data. We discuss below the architecture and the hyperparameters of the cGAN model used for this task. Given the adversarial nature of GANs, where the Generator and Discriminator engage in a Minimax game, careful tuning of key parameters is critical to ensure that both networks are well-balanced for effective training.

5.1. *Data Preparation*

First, we generate synthetic images of rapidly rotating stars by modelling them as oblate spheroids with varying radii and oblateness parameters. To incorporate the effect of gravity darkening, we also consider different viewing angles and assume a linear dependence on the declination angle of each point on the stellar surface. The traced ellipses result from integrating over the source's hour angle.

Next, Salt and Pepper noise is introduced into these images; usually this is done at the rate of 0.5% of the number of pixels in the image. Then, the images are resized and their mean is subtracted. A two-dimensional Fast Fourier Transform, along with a Fourier shift, is applied, yielding a complex number for each pixel. Since II does not measure phase, the absolute value is calculated.

In the present implementation, salt-and-pepper perturbations are applied to the stellar image before calculating its Fourier transform, and the resulting Fourier representation is supplied to the neural network. These perturbations are introduced to examine the method's response to phenomenological image corruption; they are not intended to represent photon-correlation measurement uncertainty at a prescribed II SNR. The additional perturbation used during discriminator training serves a separate computational purpose for regularization noise. Accordingly, the SNR expression derived in Section 3 should be interpreted as a theoretical description of II measurement sensitivity rather than as the noise prescription used in the current simulations. A direct assessment of observational performance requires incorporating an appropriate II correlation-noise model into the simulated measurement pipeline.

Next, sampling of the interferometric plane is introduced via pixel-wise multiplication of the absolute-valued Fourier-transformed image (left panel of Fig. 4) and the two sets of baseline tracks of (Fig. 2). It may be recalled that these two sets of baseline tracks of Fig. 2 are, respectively, generated by each of the distinct pairs of telescopes in each of the two IACT arrays of Fig. 1; i.e., the array of six IACTs (the left panel) and the array of nine IACTs (the right panel). These tracks are generated due to the Earth's rotation during the period (one night) of II observation. The result of this pixel-wise multiplication is a set of two sparsely sampled maps in the Fourier plane featuring ellipse (Fig. 3 is an example) and are shown in the middle and the right panels of Fig. 4.

Finally, we normalize the pixel values and convert them to 8-bit integers, producing an image that encodes the sparsely sampled, phase-free visibility measured by II. The image of the corresponding simulated (fictitious) star, which serves as the "ground truth" is processed identically to avoid any bias. These two images are merged side-by-side into a single image (one of is shown in Fig. 5). Thus two pairs of datasets, each corresponding to one of the two IACT arrays, are created. Each of these datasets are then split into three parts in the proportions of 80% for training, 10% for validation, and 10% as the test set. This partition of the full dataset is indicated in the Fig.6. In the following, we refer to these three parts as the Training Set, the Validation Set and the Testing Set respectively.

5.2. *GAN Architecture*

The GAN architecture used in this work is a Pix2Pix cGAN, which uses an image-to-image translation strategy with both the ground truth and the condition being images. Originally introduced by Isola et al. (Isola et al. 2017), this architecture is specifically suitable for image processing and reconstruction objectives. For instance, the TensorFlow tutorials[1] demonstrate its application to a dataset of architectural facades. This architecture has been adapted for the phase retrieval problem at hand here. The network is implemented using the TensorFlow library (Abadi et al. 2016), calculations are performed with scipy (Virtanen et al. 2020), and plots are generated with matplotlib (Hunter 2007).

5.3. *Hyperparameter Tuning*

The cGAN model architecture used in this work employs several hyper-parameters, which are explained briefly below (for a more in-depth discussion, see Murphy 2022).

The learning rate (lr) of the optimizer determines how much the model updates its parameters with each iteration. A learning rate that is too small may lead to underfitting, while one that is too large can render the model unstable. Therefore, selecting an appropriate learning rate is crucial (Murphy 2022). A canonical choice in Pix2Pix and other GAN models is $lr = 2 \times 10^{-4}$. In our case too, we found this choice to be appropriate.

The kernel size refers to the dimensions of the convolutional kernel used in the network, determining how many pixels are combined to produce a new pixel. A larger kernel size can capture features spanning several pixels, but it may also incorporate unrelated features. Small kernel sizes are preferable in cases where target images contain finer details or high spatial frequency features. Since the "ground truth" target images in our case have longer scale gravity darkening features, we have opted for the more canonical choice of kernel size being 5.

The amount of noise is controlled by two parameters, "alpha" and "beta", which indicate the percentage of pixels altered to either white or black, hence the term Salt and Pepper noise. Here, "alpha" is applied to the real image, while "beta" is applied to the generated image. Different ratios ("alpha/beta") can lead to varying model

[1] https://www.tensorflow.org/tutorials/generative/pix2pix

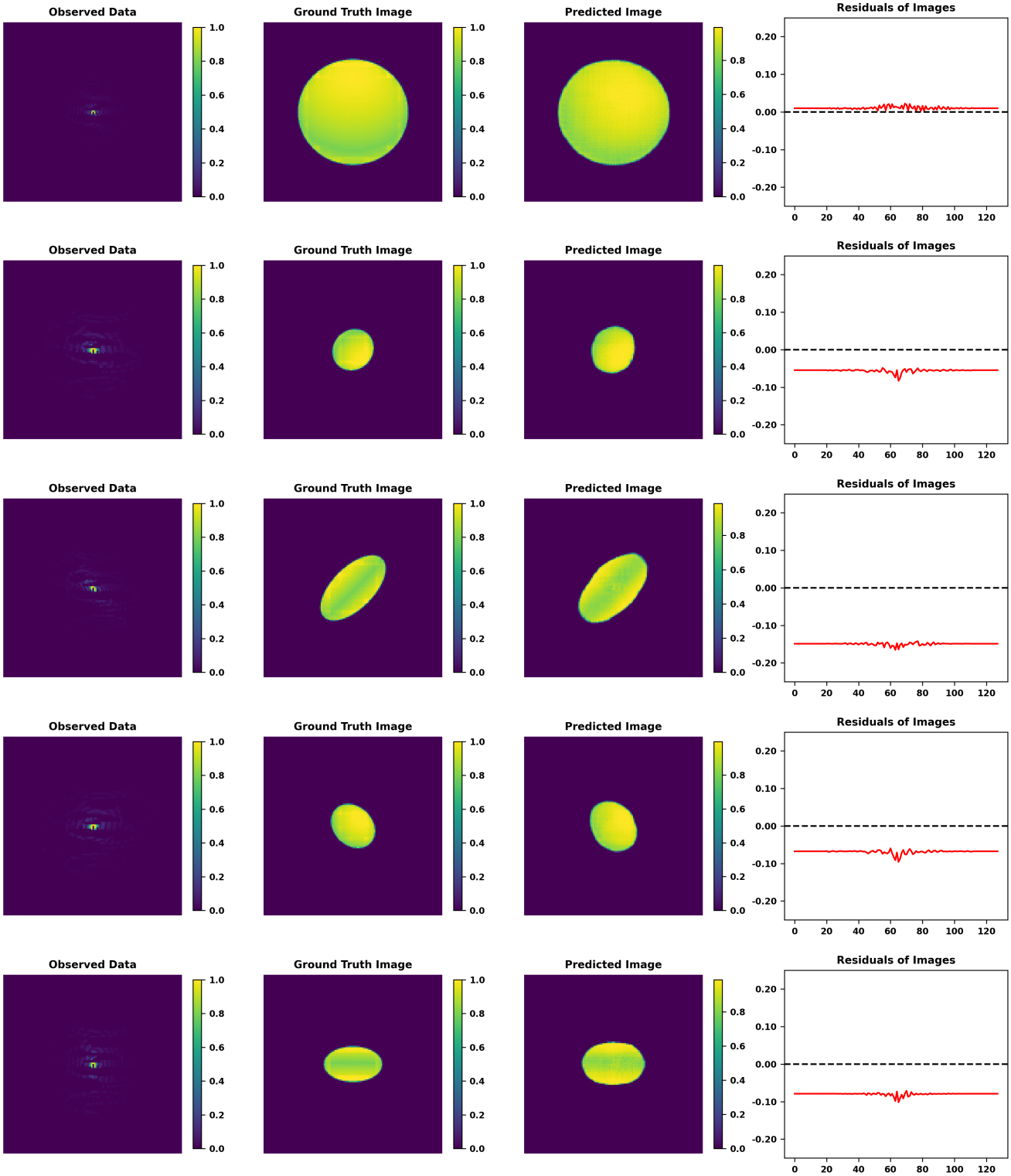


FIG. 8.— Example results of image reconstruction using the cGAN model along with the II observations simulated in this work. Each row in this figure represents the results for a hypothetical fast-rotating star. The left panel represents the sparse II pattern obtained by the simulated observation of the star using the array of $N_T = 6$ IACTs illustrated in the left panel of Fig.1. This image acts as the "condition" part of the data input to the cGAN model. The second panel from left displays the real image, or ground truth, which the Discriminator uses to distinguish from the images generated by the Generator. The data generated for training, validation and testing of the cGAN model is a merge of this image and its II pattern presented in the left panel. The third panel is the reconstructed image, or the predicted image, produced by the trained GAN model . The fourth and the last panel shows the 1D-residuals between the ground truth and the predicted image in the (u, v) plane. The x-axis of this panel represents the horizontal pixel coordinate and the y-axis shows the residual intensity value at the respective pixel.

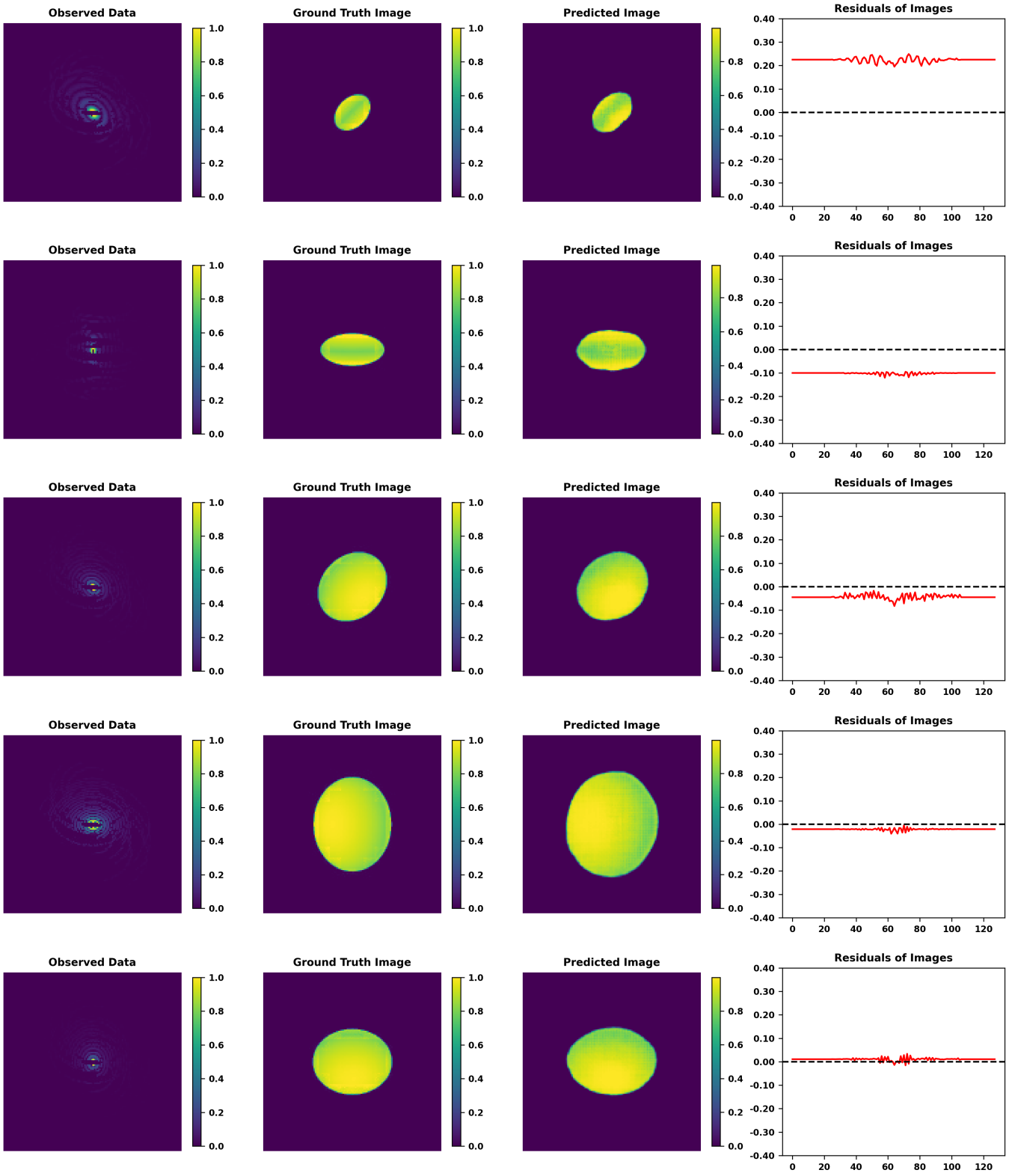


FIG. 9.— Example results of image reconstruction using the cGAN model (as presented in Fig. 8), but with the II simulations carried out with the array of $N_T = 9$ as shown in the right panel of Fig. 1. Panel labels are as described in the caption of Fig. 8.

performance; however, our results indicate that distinct noise rates do not significantly affect the loss functions.

The batch size defines the number of images processed simultaneously by the network. Smaller batch sizes have been observed to improve generalization (Prince 2023). However, because a larger batch size significantly increases training time, a batch size of 1 is used. Besides, in Pix2Pix cGAN implementations found in literature, this choice is found to be often preferred.

The buffer size of a Pix2Pix GAN refers to a small memory of image pool of previously generated images. They are occasionally fed to the Discriminator in place of the freshly generated ones. This strategy of mixing old and new fakes mitigates the risk of mode collapse wherein the Discriminator tends to map all or large number of generated images to only one or a few real "ground truth" image(s). We have chosen a fairly large buffer size (=1400) to protect the model against mode collapse.

In the training of GANs, one often-followed strategy to potentially boost performance is to give the Discriminator an advantage by increasing its number of training epochs before returning to the Generator's training. This hyperparameter is referred to as the Discriminator repetition (as seen in Table 1). While this can lower the Discriminator loss, as shown in Fig. 7, it also increases training time. In training our model, we did not notice any significant advantage derived from this strategy. Since the generated images did not noticeably improve with additional Discriminator training, we adopted the strategy of training both the networks with equal preference (Discriminator repetition = 1).

One domain specific hyperparameter of the cGAN model presented here is the Number of Telescopes N_T. The degree of sparse sampling of the intensity interferometric (II) image plane can be varied to provide the model with access to more number of active (non-zero) pixels. Point to note here, is that the coverage of the Fourier interferometric plane (number of active pixels in the II image) scales with available number of baselines, and the latter scales quadratically with the number of telescopes.

The hyperparameter Output Channels refers to the number of channels in the generator output (e.g., 1 for grayscale, 3 for RGB). It is worthwhile to recall that the cGAN model constructed in this work is trained on "ground truth" target images and the simulated II data, both in grey scales as seen in Fig.5. It is natural that the output of this model will be in grey scales only. Therefore the value of this hyperparameter is set to 1. The choice of hyperparameter λ has been commented upon earlier.

An optimized set of hyperparameters is selected through an iterative process of training and validation. For a tentatively chosen set of the hyperparameters, the model is trained using the Training Set until the both the Discriminator and the Generator loss functions are minimized. This model, thus trained along with its model parameters, is then passed through validation using the Validation Set. This cycle of training and validation is iterated till an optimal set of hypermaraters is arrived at. During each epoch of training of the model, both the Discriminator and the Generator networks are trained for 100,000 steps. Plots of the "Discriminator Loss" function and the "Generator Total Loss" function presented in Fig.7 represent the results of this training for the hyperparameters, namely, the **Discriminator** repetition ("discrep", in short). Obviously, the most compute-intensive part of this process is that of the training of the Model. The results of training and validation presented in this work were carried out on a CPU using two nodes, each with 48 threads and the entire process of training and validation required approximately 20 hours on the machine employed for this work. This iterative process of training and validation of the Model is represented schematically in the Fig.6. The chosen optimal set of hyperparameters is presented in Table-1. After training

TABLE 1
Selected hyperparameters used for training the model.

Hyperparameter	**Value**
Learning rate	2e-4
Kernel size	5
Alpha/Beta	1
Batch size	1
Buffer Size	1400
Discriminator repetition	1
Number of Telescope	6 & 9
Output Channels	1
Lambda	100

and validation, the trained network was evaluated using an independent test set comprising $N_{\rm test} = 100$ **images**. The results are presented in the following section.

The Pix2Pix cGAN architecture along with the choice of the values of the hyperparameters mentioned in the Table -1 and used in the implementation of this architecture constitutes the cGAN model (hereinafter referred to as "the GAN Model" or simply "the Model").

6. IMAGE RECONSTRUCTION: RESULTS AND EVALUATION

In this section, we examine the performance of the GAN Model whose architecture and choice of hyperparameters have been discussed above. We subject the trained Model to the task of phase retrieval and image reconstruction on the Testing Set (100 image). In this process, for each ground truth image of the Testing Set, two predicted images are generated: one each for the arrays of $N_T = 6$ and $N_T = 9$ IACTs.

6.1. *Visual Evaluation of Images Predicted by the Model*

Fig. 8 and Fig. 9 demonstrate the success of the trained Model in reconstructing the images of a sample of the fictitious fast-rotators drawn from the Testing Set.

The four panels from left to right in each row of Fig. 8 and Fig. 9 represent the following:

- The left panel represents the sparse II pattern obtained by the simulated observation using the set of IACTs illustrated in Fig.1. As explained earlier, this image acts as part of the input, namely its "condition" part. This image acts as the condition that the sky-images of the star generated by the Generator must conform to.

- The second panel from left displays the real image, or ground truth target image, which the Discriminator loss function uses to distinguish from the images generated by the Generator.

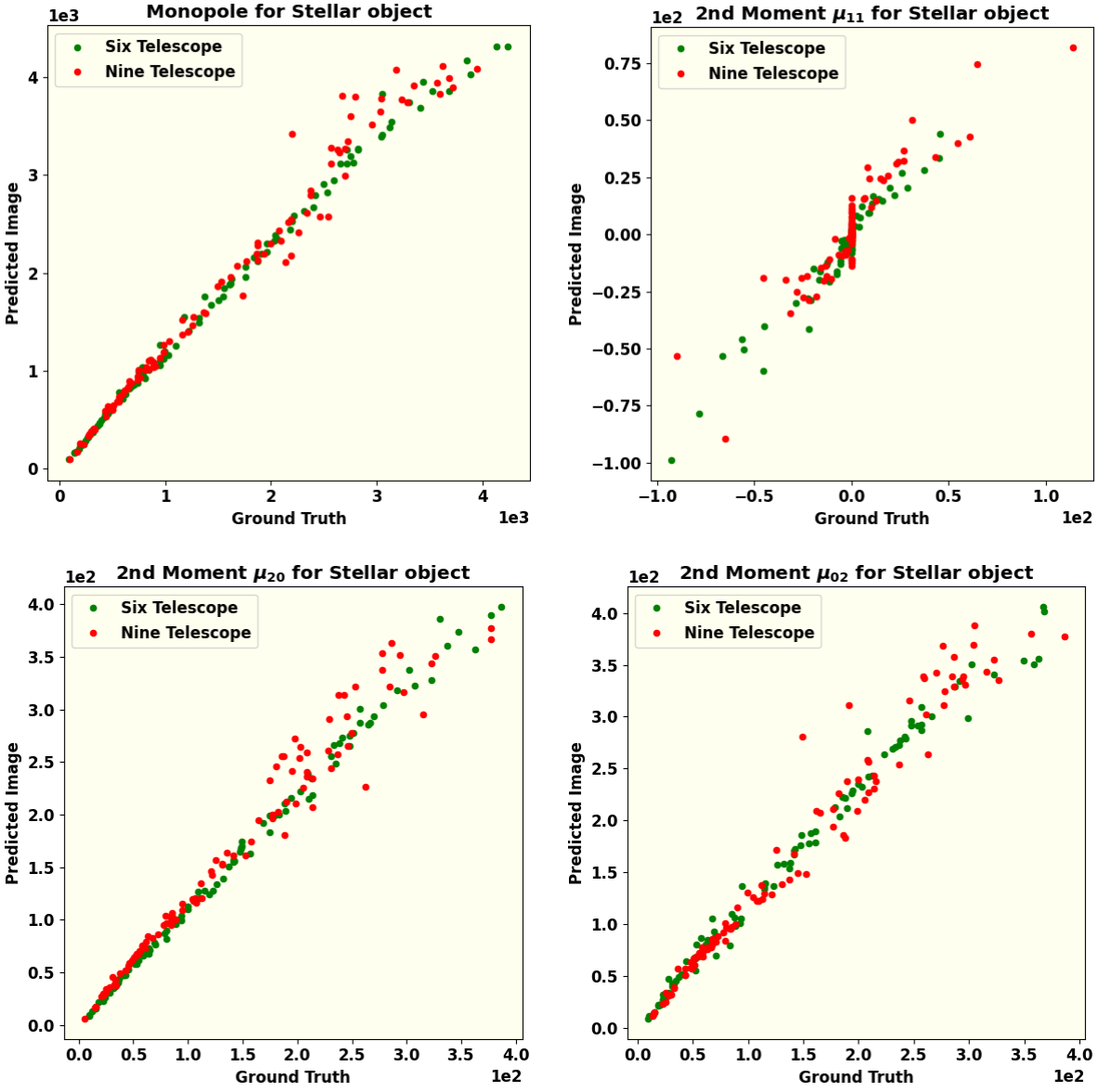


FIG. 10.— Comparison of moments of the predicted images with those of the ground-truth images. Continued on the next page.

- The third panel from the left presents the reconstructed (or predicted) image corresponding to the ground-truth image shown in the second panel and generated by the trained Model. The similarity between these two images indicates the Model's performance in its stated objective of image reconstruction.

We note that the image reconstruction problem exhibits a 180° rotational ambiguity: the reconstructed image and its 180°-rotated counterpart are physically equivalent under the symmetry considered in this work. Consequently, a direct pixel-wise comparison with the ground-truth image may penalize a reconstruction solely because of its orientation. To account for this degeneracy, we define a symmetry-aware mean squared error ($E_{\rm sym}$) as

$$E_{\rm sym} = \min\left\{\mathrm{MSE}(I_{\rm pred}, I_{\rm GT}), \mathrm{MSE}(R_{180}(I_{\rm pred}), I_{\rm GT})\right\}, \tag{24}$$

where $I_{\rm pred}$ and $I_{\rm GT}$ denote the predicted and ground-truth images, respectively, and R_{180} denotes rotation of an image by 180°. The two MSE values quantify the pixel-wise differences between the ground-truth image and the original and rotated predictions, respectively. Taking their minimum selects the orientation with the lower error, thereby evaluating the reconstruction over the two equivalent orientations. Since this selection uses the ground-truth image, $E_{\rm sym}$ represents a best-match evaluation under the assumed rotational degeneracy; it should not be interpreted as an independent recovery of the absolute orientation by the reconstruc-

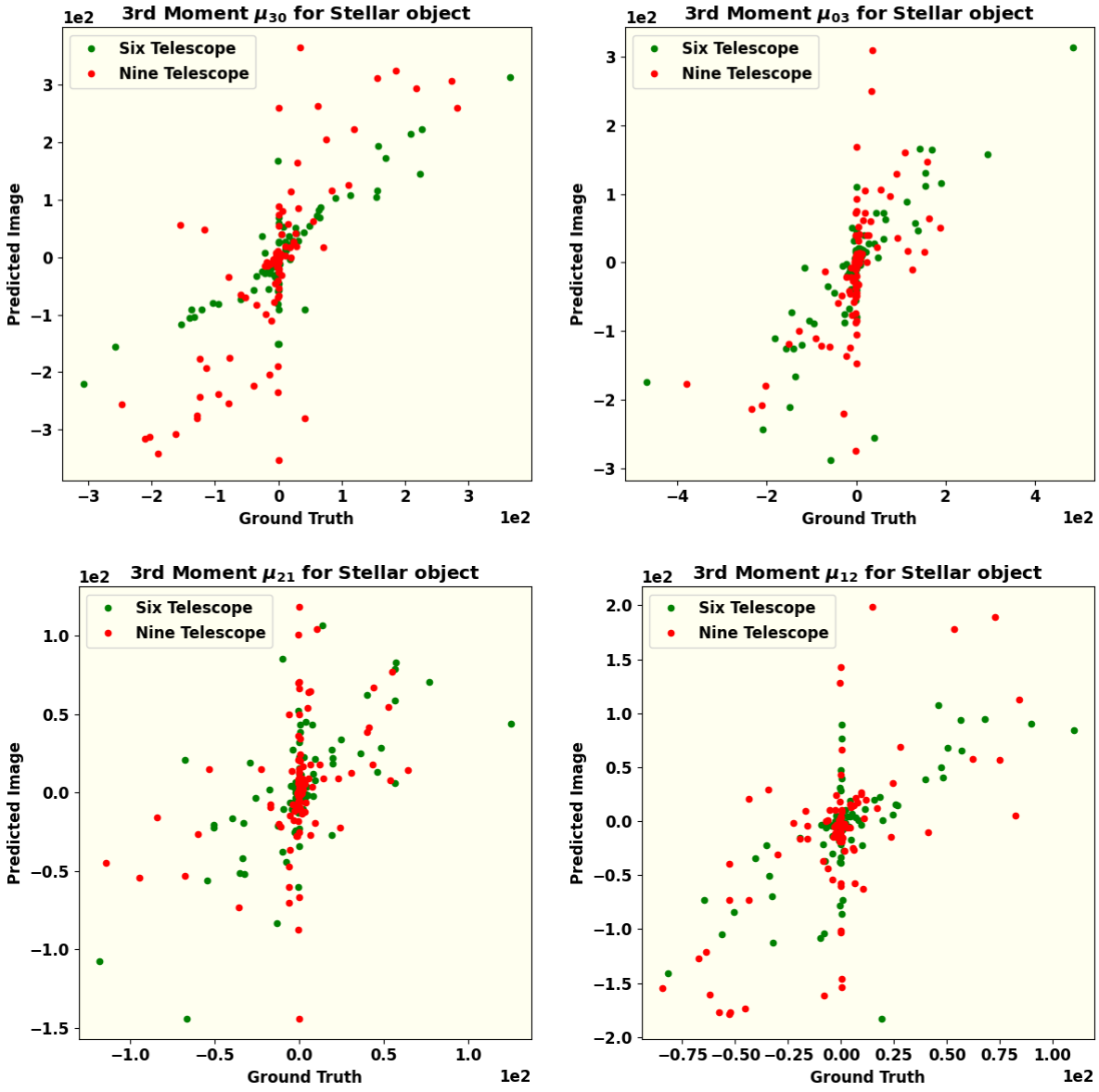


FIG. 10.— Continued. The panels of this figure depict the comparison of moments of predicted images with those of the ground-truth images. The predicted images are generated by our trained cGAN model vis. a vis. the ground-truth images in the two Test Sets. The green and red dots represent the moments of the FRSs generated from II observations with arrays of $N_T = 6$ and $N_T = 9$ IACTs, respectively. The small scatter in the moments of the predicted images indicates balanced training of the model. This figure also shows that improvement in (u, v) coverage dominates over the number of baselines.

tion model.

- The right panel shows the 1D residuals between the ground truth target image and the predicted image in the interferometric plane. The small values are indicative of the success of training the Model.

The predicted images and the residuals presented in the third and the fourth columns (from left) of Fig. 8 and Fig. 9 show encouraging results, conveying visual information about the source's size, shape, and brightness distribution across its surface fairly accurately. This has been achieved on the basis of the input provided by II observation using only 15 and 36 baselines, corresponding to six and nine telescopes, respectively. Further improvements could surely be achieved by increasing the number of telescopes to enhance the coverage of the (u, v) plane. A closer examination of this proposition might be able to contribute to the design and instrumentation aspects in the existing and upcoming CTAO.

6.2. *Evaluation of the Model using Moments*

The reconstructed images are visually compelling, demonstrating the Model's effectiveness in using II to reconstruct images. However, visual assessment alone is insufficient; statistical evaluation of the generated images in comparison with the ground truth images is necessary to validate the results. We present here the results of our calculation and comparison of the moments of distribution of pixel brightness in the target "Ground Truth" images and the "Predicted" images.

Image moments capture key properties of the reconstructed objects, such as shape, size, and intensity dis-

TABLE 2
Scatter metrics of the reconstructed image sets relative to the ground-truth images. The $N6$ and $N9$ sets correspond to the six- and nine-telescope configurations, respectively. The predicted images are evaluated after selecting, for each test example, the orientation (original or rotated by 180°) that minimizes the MSE relative to the corresponding ground truth. Thus, the reported metrics represent best-match agreement under the assumed rotational degeneracy. Here, S_c denotes the centroid scatter, and S_{pq} denotes the scatter of the central moments of order $p+q$.

Predicted Set	S_c (Norm. Inten.)	$S_{11}(\mathrm{mas})^2$	$S_{20}(\mathrm{mas})^2$	$S_{02}(\mathrm{mas})^2$	$S_{12}(\mathrm{mas})^3$	$S_{21}(\mathrm{mas})^3$	$S_{30}(\mathrm{mas})^3$	$S_{03}(\mathrm{mas})^3$
N6	0.0223 ±	0.387 ±	7.606 ±	7.296 ±	35.822 ±	30.603 ±	102.691 ±	91.102 ±
N9	0.0256 ±	0.499 ±	11.446 ±	10.869 ±	36.159 ±	35.865 ±	108.123 ±	108.622 ±

tribution, by quantifying features like position, orientation, and brightness distribution. By comparing the moments of the Model-generated images with those of the ground truth, we can objectively assess the consistency and accuracy of the reconstruction. This approach provides a reliable framework for evaluating reconstruction quality, as image moments can reveal subtle differences in geometric and intensity properties that might not be apparent through visual inspection alone.

The raw moment M_{ij} of an image is defined as (Hu 1962)

$$M_{ij} = \sum_x \sum_y x^i y^j I(x,y) \tag{25}$$

where $I(x,y)$ represents the intensity at pixel (x,y) in mas. The zeroth order raw moment, or monopole, represents the total intensity of an image. It is computed by summing all pixel values across the image, yielding an overall intensity measure. In this context, analyzing the monopole provides the total pixel brightness of the images of the fictitious stars. According to Eq. (25), the monopole of an image is calculated as:

$$M_{00} = \sum_x \sum_y I(x,y). \tag{26}$$

The first order raw moments normalized by the respective monopole moments of the images given by

$$x_c = \frac{M_{10}}{M_{00}} = \frac{\sum_{x,y} x \cdot I(x,y)}{\sum_{x,y} I(x,y)}$$
$$y_c = \frac{M_{01}}{M_{00}} = \frac{\sum_{x,y} y \cdot I(x,y)}{\sum_{x,y} I(x,y)} \tag{27}$$

represent the centroids (x_c, y_c) of the pixel brightness distribution of the images in mas.

Furthermore, these calculated centroids are instrumental in analyzing the shape, size, and brightness distribution of the stars using higher-order image moments. To this end, the central moment of an image is calculated according to:

$$\mu_{pq} = \frac{1}{M_{00}} \sum_x \sum_y (x - x_c)^p (y - y_c)^q I(x,y). \tag{28}$$

The sum of p and q defines the order of the central moment.

The second order $(p+q=2)$ central moment (mas^2) of the brightness distribution of the images is analogous to the moment of inertia of a mass distribution.

Fig. 10 presents the comparison of monopole, second-order central moments $(\mu_{11}, \mu_{20}, \mu_{02})$, and third-order central moments in mas^3 $(\mu_{30}, \mu_{03}, \mu_{21}, \mu_{12})$ which are used to study the shape, structure and brightness distribution of a fast-rotating star along the line of sight, respectively using 6- and 9- IACT facilities (as explained in the upcoming subsection). The figure also highlights the impact of (u,v)-plane coverage on image reconstruction.

The green data points correspond to the monopole and moments obtained with six telescopes arranged as shown in the left panel of Fig. 1, while the red points represent results from the nine-telescope configuration displayed in the right panel of the same figure. The red points exhibit a comparatively larger scatter than the green ones, suggesting that improved (u,v)-plane coverage plays a more significant role than simply increasing the number of baselines.

A numerical comparison of scatter of the moments of the reconstructed (or predicted) images with reference to the ground truth images is worked out and presented here. In order to evaluate the "scatter metrics" of the moments, let us first extend the notations given in the formulas of raw moments (as in eq.(25), the centroids (as in eqs.(27) and the central moments (as in eq.(28) and apply them to the three sets of images produced in the testing process. Thus we define the raw moments M_{ij}^{GT}, M_{ij}^{N6} and M_{ij}^{N9} for each image belonging to the set of ground truth ("GT"), and for the reconstructed "N6", "N9" image sets respectively. Extending the same logic, we use the eqs.(27) to define the centroids $\left\{\left(x_c^{\mathrm{GT}}, y_c^{\mathrm{GT}}\right), \left(x_c^{\mathrm{N6}}, y_c^{\mathrm{N6}}\right), \left(x_c^{\mathrm{N9}}, y_c^{\mathrm{N9}}\right)\right\}$ respectively for each of the images in the three data sets. Once again, using eqn. (28), for each image in each of the datasets, we can define the central moments.

Now, the scatter of the centroids S_c^{N6} for the array of $N_T = 6$ IACT is given by

$$S_c^{\mathrm{N6}} = \frac{1}{N} \sqrt{\sum_{i=1}^{N} \left[(x_{c,i}^{\mathrm{GT}} - x_{c,i}^{\mathrm{N6}})^2 + (y_{c,i}^{\mathrm{GT}} - y_{c,i}^{\mathrm{N6}})^2\right]} \tag{29}$$

where the summation goes over all the images in the two sets. Similarly, the scatter S_{pq}^{N6} of the central moments of the N6 set of predicted images with reference to the ground truth images can be defined as

$$S_{pq}^{\mathrm{N6}} = \frac{1}{N} \sqrt{\sum_{i=1}^{N} \left(\mu_{(pq),i}^{\mathrm{GT}} - \mu_{(pq),i}^{\mathrm{N6}}\right)^2} \tag{30}$$

where the order parameter of the central moments is decided by the set of integer values $\{p, q = 0, 1, 2, 3\}$ and the order of the central moment being the the sum of p and q. The scatter metrics S_c^{N9} and S_{pq}^{N9} for the predicted images of the N9 set can also be defined and calculated using formulas similar to eqs.(29)

and (30). Table 2 summarizes these scatter metrics between the ground-truth and reconstructed images for the centroids of the images and each central moment. **The results in Table 2 also show that the 6T configuration yields smaller scatter than the 9T configuration across the reported metrics for the cases considered. However, the two configurations differ not only in telescope number but also in array geometry and the resulting Fourier-plane sampling. Consequently, this comparison does not isolate the effect of baseline number from that of spatial-frequency coverage. The observed differences should therefore be interpreted as configuration-dependent performance rather than evidence that a smaller telescope array is intrinsically superior. A controlled investigation in which array geometry and Fourier-plane coverage are varied systematically would be needed to quantify the relative contributions of these factors.**

Overall, the close agreement between the ground truth and predicted moments demonstrates the effectiveness of the model in reconstructing images using II. The relatively small scatter in the predicted moments further indicates that the model has learned the underlying features robustly, without significant over-fitting.

6.3. *The reconstructed Parameters for object*

The centroids (x_c, y_c) indicate only the center of the star and its spatial location in the image. In contrast, the second-order central moments determine the orientation, semi-major axis, and eccentricity relative to the source's center (Teague 1980). These moment-based parameters fully describe the two-dimensional ellipse that fits the image data.

The position angle of the stellar image in the plane of the sky can be determined from the second-order central moments as

$$\theta = \tfrac{1}{2} \arctan\left(\frac{2\mu_{11}}{\mu_{20} - \mu_{02}}\right). \tag{31}$$

The semi-major and semi-minor axes of the stellar object are computed using the second-order central moments and are denoted as a and b, respectively.

$$\begin{aligned} a &= 2\sqrt{mp + \delta} \\ b &= 2\sqrt{mp - \delta} \end{aligned} \tag{32}$$

where,

$$mp = \frac{\mu_{20} + \mu_{02}}{2} \tag{33}$$

and

$$\delta = \frac{\sqrt{4\mu_{11}^2 + (\mu_{20} - \mu_{02})^2}}{2}. \tag{34}$$

Using the calculated axis values, **the eccentricity of the fast-rotating star is determined as**:

$$e = \sqrt{1 - \frac{b^2}{a^2}}. \tag{35}$$

These parameters (Eqs. 31–35) characterize the overall geometry of the stellar intensity distribution. In particular, the semi-major and semi-minor axes describe its spatial extent, while the eccentricity quantifies its degree of elongation. The position angle specifies the orientation of the major axis in the image plane. Furthermore, the brightness distribution and its asymmetry can be examined using third- and higher-order image moments.

To assess the reconstruction performance, we compare the image moments of the reconstructed images with those of the corresponding ground-truth (GT) images for all 100 test cases. The relationships between the moments of the GT, N6, and N9 images are shown in Fig. 10. These comparisons provide a visual assessment of how well the reconstructed images reproduce the spatial distribution and overall morphology of the reference stellar images. In particular, the second-order moments characterize the recovery of the source extent and elongation, whereas the higher-order moments provide information about the reconstructed brightness distribution.

In addition to the moment-based comparison, we evaluate the errors in the geometric parameters defined by Eqs. 31–35 for the 100 test images. Table 3 summarizes the mean errors and their standard deviations for the N6 and N9 reconstructions. This statistical comparison provides a quantitative assessment of the accuracy and consistency with which the two configurations recover the geometric properties of the reference stellar intensity distributions.

TABLE 3
Mean absolute errors (MAE) and their standard deviations for the reconstructed geometric parameters of the 100 test images for the N6 and N9 configurations. The errors are calculated with respect to the corresponding ground-truth (GT) images. The 180° ambiguity in the position angle is taken into account.

Parameter	N6 (Mean ± Std.)	N9 (Mean ± Std.)
Semi-major axis, a	—	—
Semi-minor axis, b	—	—
Position angle, θ (°)	—	—
Eccentricity, e	—	—

7. CONCLUSION

Intensity interferometry (II) is re-emerging as a promising technique for optical long-baseline interferometry. However, unlike conventional amplitude interferometry, II measures the correlation of photon arrival times and provides information on the squared visibility, resulting in the loss of phase information. Recovering the source morphology from such measurements therefore remains an important challenge.

In this work, we have explored the feasibility of using a conditional Generative Adversarial Network (cGAN), based on the Pix2Pix framework, for image reconstruction from II data. The model was trained using paired synthetic stellar images and their corresponding II data, with the ground-truth sky images serving as targets and the II measurements as conditional inputs. The generator learns to reconstruct images consistent with the input II data, while the discriminator encourages the generation of realistic stellar intensity distributions. The re-

construction performance was evaluated by comparing the image moments of the generated and ground-truth images, including the monopole and second- and third-order moments. The results demonstrate that the proposed approach can recover the overall morphology and brightness distribution of the synthetic rapidly rotating stars considered in this study.

The present work provides a controlled proof of concept rather than a comprehensive assessment of image reconstruction for realistic stellar surfaces. The adopted synthetic images have smooth brightness distributions and do not incorporate features such as starspots, granulation, realistic limb darkening, or tidal deformation in interacting binary systems. Moreover, the adopted brightness prescription is phenomenological and should not be interpreted as a complete physical model of gravity darkening. Thus, the demonstrated reconstruction capability is limited to the synthetic benchmark investigated here.

Several extensions are required to establish the applicability of this approach to more realistic astrophysical sources. Future work should incorporate physically motivated gravity-darkening prescriptions, wavelength-dependent limb darkening, surface inhomogeneities, and tidal distortions. It will also be important to investigate the effects of Fourier-plane coverage, telescope-array geometry, observing time, and detector efficiency on reconstruction performance. In particular, the improved spatial-frequency sampling potentially offered by larger IACT arrays and future facilities such as CTAO may provide valuable opportunities to investigate more complex stellar intensity distributions. The effects of these factors will need to be assessed through systematic simulations.

Although the relatively simple sources considered here could also be investigated using conventional parameter-fitting techniques, the objective of this study was to explore machine-learning-based image reconstruction from sparsely sampled II data. The present results do not establish the superiority of cGAN over conventional parameter-fitting methods, nor do they quantify the specific contribution of adversarial learning relative to an L1-only U-Net. Comparisons with these approaches, along with investigations of alternative network architectures and loss functions, will be necessary to assess their relative advantages and limitations.

Overall, this study represents an initial step toward applying machine-learning techniques to image reconstruction in II. Further development and validation using more realistic stellar models, improved noise descriptions, and different telescope-array configurations will help determine the potential of this approach for studying the morphology and surface-brightness distributions of rapidly rotating and interacting stars.

NOTE ON SOFTWARE

The code used for this work is available on the DOI: 10.5281/zenodo.17598807

ACKNOWLEDGEMENTS

We sincerely thank the anonymous referee for the constructive comments and valuable suggestions, which have significantly improved the quality of this manuscript. We also acknowledge the Padmanabha Cluster at IISER Thiruvananthapuram and the Surya Cluster at ARIES for providing the computational facilities and resources required for this work. One of the authors (SS) gratefully acknowledges the computing facilities and the local hospitality extended to him by the Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India under its Visiting Associate Programme during the preparation and finalization of this manuscript.

REFERENCES

Abadi, M., Agarwal, A., Barham, P., et al. 2016, arXiv preprint arXiv:1603.04467, doi: 10.48550/arXiv.1603.04467

Abe, S., Abhir, J., Acciari, V., et al. 2024, MNRAS, 529, 4387, doi: 10.1093/mnras/stae697

Acciari, V., Bernardos, M., Colombo, E., et al. 2020, MNRAS, 491, 1540, doi: 10.1093/mnras/stz3171

Acharyya, A., Aufdenberg, J., Bangale, P., et al. 2024, The Astrophysical Journal, 966, 28, doi: 10.3847/1538-4357/ad2b68

Aleksić, J., Ansoldi, S., Antonelli, L. A., et al. 2016, Astroparticle Physics, 72, 76, doi: 10.1016/j.astropartphys.2015.02.005

Archer, A., Aufdenberg, J. P., Bangale, P., et al. 2025, ApJ, 995, 191, doi: 10.3847/1538-4357/ae0744

Baumgartner, S., Bernardini, M., Canivete Cuissa, J. R., et al. 2020, MNRAS, 498, 4577, doi: 10.1093/mnras/staa2638

Brown, R. H., & Twiss, R. Q. 1957, Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences, 242, 300. https://www.jstor.org/stable/100432

—. 1958, Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences, 243, 291. https://www.jstor.org/stable/100349

Coccomini, D., Messina, N., Gennaro, C., & Falchi, F. 2021, arXiv preprint arXiv:2111.11578, doi: 10.48550/arXiv.2111.11578

Domiciano de Souza, A., Kervella, P., Jankov, S., et al. 2003, Astronomy and Astrophysics, 407, L47, doi: 10.1051/0004-6361:20030868

—. 2005, Astronomy and Astrophysics, 442, 567, doi: 10.1051/0004-6361:20042476

Dravins, D., LeBohec, S., Jensen, H., et al. 2013, Astropart. Phys., 43, 331, doi: 10.1016/j.astropartphys.2012.04.017

Fienup, J. 1982, Appl. Opt., 21, 2758, doi: 10.1364/AO.21.002758

Gamo, H. 1963, J. Appl. Phys., 34, 875, doi: 10.1063/1.1729553

Gerchberg, R. W. 1972, Optik, 35, 237

Glauber, R. J. 1963, Physical Review, 130, 2529, doi: 10.1103/PhysRev.130.2529

Goldberger, M. L., Lewis, H. W., & Watson, K. M. 1963, Phys. Rev, 132, 2764, doi: 10.1103/PhysRev.132.2764

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., et al. 2014, NeurIPS, 27, doi: 10.48550/arXiv.1406.2661

Hanbury Brown, R., Davis, J., & Allen, L. 1974, Monthly Notices of the Royal Astronomical Society, 167, 121, doi: 10.1093/mnras/167.1.121

Hanbury Brown, R., & Twiss, R. Q. 1956, Nature, 177, 27, doi: 10.1038/177027a0

Haubois, X., Perrin, G., Lacour, S., et al. 2009, A & A, 508, 923, doi: 10.1051/0004-6361/200912927

Holmes, R. B., Lebohec, S., & Nunez, P. D. 2010, in Adaptive Coded Aperture Imaging, Non-Imaging, and Unconventional Imaging Sensor Systems II, ed. J. J. Dolne, T. J. Karr, V. L. Gamiz, S. Rogers, & D. P. Casasent, Vol. 7818, International Society for Optics and Photonics, 78180O, doi: 10.1117/12.858438

Hu, M.-K. 1962, IRE transactions on information theory, 8, 179, doi: 10.1109/TIT.1962.1057692
Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55
Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. 2017, in Proceedings of the IEEE conference on computer vision and pattern recognition, 1125–1134, doi: 10.48550/arXiv.1611.07004
Kirisits, C., Raik, K., Scherzer, O., Strohmenger, C., & Yan, J. 2024, arXiv preprint arXiv:2406.14143v2, doi: 10.48550/arXiv.2406.14143
Le Bohec, S; Holder, J. 2006, The Astrophysical Journal, 649, 399, doi: 10.1086/506379
Li, X., Gao, X., Tang, J., et al. 2014, in Imaging Spectroscopy, Telescopes and Large Optics, Vol. 9298, SPIE, 92981G1–G7, doi: 10.1117/12.2072253
Liu, L.-C., Wu, C., Li, W., et al. 2024, arXiv preprint, arXiv:2404.15685, doi: 10.48550/arXiv.2404.15685
—. 2025, PRL, 134, 180201, doi: 10.1103/PhysRevLett.134.180201
Lucy, L. B. 1967, Zeitschrift für Astrophysik, 65, 89. https://ui.adsabs.harvard.edu/abs/1967ZA.....65...89L/abstract
Maeder, A. 1999, Astronomy & Astrophysics, 347, 185. https://ui.adsabs.harvard.edu/abs/1999A&A...347..185M
Martinez, A. O., Baron, F., Monnier, J. D., Roettenbacher, R. M., & Parks, J. R. 2021, The Astrophysical Journal, 916, 60, doi: 10.3847/1538-4357/ac06a5
McAlister, H. A., Ten Brummelaar, T., Gies, D., et al. 2005, The Astrophysical Journal, 628, 439, doi: 10.1086/430730
Mirza, M., & Osindero, S. 2014, arXiv preprint arXiv:1411.1784, doi: 10.48550/arXiv.1411.1784
Monnier, J. D., Zhao, M., Pedretti, E., et al. 2007, Science, 317, 342, doi: 10.1126/science.1143205
Murphy, K. P. 2022, Probabilistic machine learning: an introduction (MIT press)
Mustafa, M., Bard, D., Bhimji, W., et al. 2019, Computational Astrophysics and Cosmology, 6, 1, doi: 10.1186/s40668-019-0029-9
Norris, R. P., Baron, F. R., Monnier, J. D., et al. 2021, ApJ, 919, 124, doi: 10.3847/1538-4357/ac0c7e
Nuñez, P. D., LeBohec, S., Kieda, D., et al. 2010, in Optical and Infrared Interferometry II, Vol. 7734, SPIE, 458–467, doi: 10.1117/12.856496
Nuñez, P. D., Holmes, R., Kieda, D., & LeBohec, S. 2012a, Monthly Notices of the Royal Astronomical Society, 419, 172, doi: 10.1111/j.1365-2966.2011.19683.x
Nuñez, P. D., Holmes, R., Kieda, D., Rou, J., & LeBohec, S. 2012b, Monthly Notices of the Royal Astronomical Society, 424, 1006, doi: 10.1111/j.1365-2966.2012.21263.x
Pedretti, E., Monnier, J. D., ten Brummelaar, T., & Thureau, N. D. 2009, New Astronomy Reviews, 53, 353, doi: 10.1016/j.newar.2010.07.005
Prince, S. J. 2023, Understanding deep learning (MIT press)
Rai, K. N., Basak, S., & Saha, P. 2021, MNRAS, 507, 2813, doi: 10.1093/mnras/stab2391
Rai, K. N., Basak, S., Sarangi, S., & Saha, P. 2025, Resonance, 30, 45, doi: 10.1007/s12045-025-1729-x
Rai, K. N., Sarangi, S., Saha, P., & Basak, S. 2022, MNRAS, 516, 2864, doi: 10.1093/mnras/stac2433
Ronneberger, O., Fischer, P., & Brox, T. 2015, in Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18, Springer, 234–241, doi: 10.1007/978-3-319-24574-4_28
Sato, T., Ishii, J., & Wadaka, S. 1979, Appl. Opt., 18, 485, doi: 10.1364/AO.18.000485
Sato, T., Sasaki, K., & Ando, K. 1981, Appl. Opt., 20, 2055, doi: 10.1364/AO.20.002055
Sato, T., Wadaka, S., Yamamoto, J., & Ishii, J. 1978, Appl. Opt., 17, 2047, doi: 10.1364/AO.17.002047
Schawinski, K., Zhang, C., Zhang, H., Fowler, L., & Santhanam, G. K. 2017, MNRAS, 467, L110, doi: 10.1093/mnrasl/slx008
Scuderi, S., Giuliani, A., Pareschi, G., et al. 2022, Journal of High Energy Astrophysics, 35, 52, doi: 10.1016/j.jheap.2022.05.001
Teague, M. R. 1980, Josa, 70, 920, doi: 10.1364/JOSA.70.000920
—. 1983, Journal of Opticala Society of America, 73, 1434, doi: 10.1364/JOSA.73.001434
van Belle, G. T., Ciardi, D. R., Thompson, R. R., Akeson, R. L., & Lada, E. A. 2001, The Astrophysical Journal, 559, 1155, doi: 10.1086/322340
Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature methods, 17, 261, doi: 10.1038/s41592-019-0686-2
Vogel, N., Zmija, A., Wohlleben, F., et al. 2025, Monthly Notices of the Royal Astronomical Society, 537, 2334, doi: 10.1093/mnras/stae2643
Von Zeipel, H. 1924, Monthly Notices of the Royal Astronomical Society, Vol. 84, p. 665-683, 84, 665, doi: 10.1093/mnras/84.9.665
Zhang, J., Chen, Q., Sun, J., Tian, L., & Zuo, C. 2020, Opt. Lett., 45, 3649, doi: 10.1364/OL.391823
Zhao, M., Monnier, J. D., Pedretti, E., et al. 2009, The Astrophysical Journal, 701, 209, doi: 10.1088/0004-637X/701/1/209

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